

# Induced-Gravity Cosmology: Inflation, Dark Energy, and Dark Matter from a Common Scalar Origin

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## Abstract

We construct a scalar-tensor cosmology within pure induced gravity, in which a single real scalar field  $\Phi$  with a  $\mathbb{Z}_2$ -symmetric potential  $V_J(\Phi) = \frac{\lambda_0}{4}(\Phi^2 - \Phi_0^2)^2 + V_c$  and non-minimal coupling  $\xi\Phi^2 R$  replaces the Einstein–Hilbert term, generating the Planck mass via  $\langle\Phi\rangle = M_{\text{Pl}}/\sqrt{\xi}$ . Conformal transformation to the Einstein frame yields a Starobinsky-type slow-roll plateau. With locked  $N \equiv \ln(a_{\text{end}}/a_*)$  and exact potential slow-roll,  $\xi = 11.1$  gives  $N = 50 \Rightarrow \lambda_0 \simeq 6.70 \times 10^{-8}$ ,  $r \simeq 0.00425$ ,  $n_s \simeq 0.962$ ; anomaly  $T_{\text{reh}} \simeq 10^9$  GeV maps to  $N \simeq 51$ . Planck  $2\sigma$  selects  $N \approx 45\text{--}56$  ( $r \leq 0.0052$ ); LiteBIRD  $r > 0.01$  would exclude the model class. A Dirac fermion  $\psi$  with  $-g\Phi\bar{\psi}\psi$  is a cold-dark-matter *candidate*:  $m_\psi = g\Phi_0$  shares the same VEV; stability uses a discrete  $\mathbb{Z}_2^\psi$  symmetry and a conformally vanishing  $\chi\psi\bar{\psi}$  vertex (both structural consequences of  $F = \xi\Phi^2$ ), conditional on an anomaly-free UV completion. Matching  $\Omega_{\text{DM}}$  with the exact mode-equation spectrum at the end-of-inflation transition selects  $g \simeq 1.0 \times 10^{-7}$  ( $m_\psi \simeq 7.5 \times 10^{10}$  GeV  $\simeq 4.6 \times 10^{-3} H_{\text{inf}}$ ), cold kinematics ( $\lambda_{\text{fs}} \sim 10^{-19}$  Mpc); absolute normalization awaits a lattice computation. The potential minimum furnishes a frozen cosmological constant with  $w = -1$  (residual-condensate corrections  $\leq 10^{-10}$ ); **this dark-energy sector is calibrated, not predicted**, with  $V_c$  fixed by  $\Omega_\Lambda = 0.683$  and the  $10^{110}$  hierarchy unexplained. The identity  $\Omega = \Phi/\Phi_0$  decouples  $\chi$  from  $\psi$  at tree level; large  $m_\chi \sim 10^{13}$  GeV suppresses SM fifth forces. Tree-level  $\chi \rightarrow h\bar{h}$  and  $\chi \rightarrow \psi\bar{\psi}$  are closed; reheating is via conformal anomaly. The SM embedding is RG-stable ( $\Delta\xi \leq 10^{-6}$ ); the late-time  $w_0 = -1$  prediction is shared with  $\Lambda\text{CDM}$  and is therefore not independently discriminating, so the sharp falsification handles remain the inflationary observables.

**Keywords:** induced gravity, scalar–tensor cosmology, Starobinsky plateau, gravitational particle production, superheavy dark matter, cosmological constant

# 1 Introduction

Modern  $\Lambda$ CDM cosmology achieves remarkable observational success, yet the physical origins of its key components—inflation, dark matter, and dark energy—remain unresolved. The Einstein–Hilbert term is treated as fundamental, and the three sectors are described by independent mechanisms with no shared microphysics.

Induced gravity [1, 2] offers an elegant conceptual simplification: the Planck mass is not fundamental but emerges from the vacuum expectation value of a real scalar field  $\Phi$ . The Einstein–Hilbert term  $M_{\text{Pl}}^2 R/2$  is replaced by  $\xi \Phi^2 R/2$ ; without field condensation, there is no spacetime gravity. At low energies, General Relativity is recovered as  $\Phi \rightarrow M_{\text{Pl}}/\sqrt{\xi}$ .

The Einstein-frame potential is shown in Fig. 1. The domain-wall / EFT-boundary structure at  $\Phi = 0$  is illustrated in Fig. 3. At large field values, the non-minimal coupling produces, after conformal transformation to the Einstein frame, a Starobinsky-type slow-roll plateau [3, 4]. The plateau geometry follows unavoidably from the same coupling that generates  $M_{\text{Pl}}$ .

Once this step is taken, a single field  $\Phi$  with a single  $\mathbb{Z}_2$ -symmetric potential  $V_J(\Phi)$  can simultaneously account for:

- **The Planck scale:**  $\langle \Phi \rangle = M_{\text{Pl}}/\sqrt{\xi}$  gives the observed gravitational coupling.
- **Inflation:** at  $\Phi \gg \Phi_0$ , conformal transformation yields a Starobinsky-type plateau. Attractor forms  $n_s \simeq 1 - 2/N$  and  $r \simeq 8/(\beta^2 N^2)$  are analytic guides; all numerical values in this paper use the locked cosmological definition  $N = \ln(a_{\text{end}}/a_*)$  with exact potential slow-roll (Sec. III), which at  $N = 50$  gives  $r \approx 0.00425$  rather than the attractor 0.00487.
- **Dark matter:** a Dirac fermion  $\psi$ , with mass  $m_\psi = g\Phi_0$  set by the same VEV, is gravitationally produced at the end of inflation and is stable under the discrete  $\mathbb{Z}_2^\psi$  symmetry together with the conformally vanishing  $\chi\psi\bar{\psi}$  vertex; the condensate itself decays to complete reheating. The relic abundance requires a preheating simulation (Sec. V).
- **Dark energy:** the residual vacuum energy  $V_c$  at  $\Phi = \Phi_0$  gives  $w = -1$  (up to residual-condensate corrections; App. D5), *calibrated empirically* to  $\Omega_\Lambda$  — not a dynamical prediction of residual inflaton motion.
- **Fifth-force screening:** the conformal identity  $\Omega = \Phi/\Phi_0$  (forced by  $F = \xi\Phi^2$ ) decouples the scalar from the dark-matter fermion  $\psi$  at tree level; for Standard Model matter, the scalar’s large mass  $m_\chi \sim 10^{13}$  GeV (Yukawa range  $\sim 10^{-27}$  cm) suppresses fifth forces, automatically satisfying solar-system bounds.
- **$\mathbb{Z}_2$  domain wall:** the surface  $\Phi = 0$  is a classical EFT boundary where  $F(0) = F'(0) = 0$  renders the Israel junction conditions inconsistent.

**Relation to existing work.** Induced gravity was originally proposed as an alternative to the Einstein–Hilbert action [1, 2] and later recognized as a natural framework for inflation [5, 6]. Recent work has explored induced-gravity inflation with various potentials [7, 8], while the non-minimal coupling  $\xi\Phi^2 R$  is shared with Higgs inflation [3] (though at  $\xi \sim 10^4$  rather than  $\xi \sim 11.1$ ).  $R^2$  (Starobinsky) inflation [4] produces identical  $n_s(N)$  and  $r(N)$  functional forms but lacks the unified dark-energy/dark-matter embedding. The idea of unifying inflation and dark energy through a single

scalar potential has been explored in quintessential inflation [9, 10], typically with disparate scales bridged by tracker or scaling solutions. The present work differs in three respects: (a) pure induced gravity (no independent  $M_{\text{Pl}}^2 R/2$  term) with the  $\mathbb{Z}_2$ -symmetric double-well potential simultaneously provides the inflationary plateau, the dark-energy vacuum energy, and the dark-matter mass scale from a single VEV; (b) the conformal identity  $\Omega = \Phi/\Phi_0$ —a structural consequence of  $F = \xi\Phi^2$ —closes the perturbative Yukawa reheating channel and decouples the dark-matter fermion at tree level, making gravitational particle production the sole DM source; (c) the  $\chi \rightarrow h\bar{h}$  tree-level decay is structurally closed in induced gravity (in contrast to  $R^2$  inflation), making the conformal-anomaly decay the dominant reheating channel. These features are not imposed by hand but follow unavoidably from the induced-gravity coupling.

This paper constructs this unified framework. Section II lays the theoretical foundation. Sections III–IV cover inflation and reheating. Section V treats the gravitationally produced fermion  $\psi$  as the dark matter candidate. Section VI derives the frozen  $\Lambda$ CDM dark energy. Section VII presents the active order parameter. Section VIII embeds the Standard Model and verifies RG stability. Section IX analyzes the  $\mathbb{Z}_2$  domain wall as a classical EFT boundary. Section X synthesizes the unified picture. Section XI collects falsifiable predictions. Section XII enumerates open problems.

We define the coupling function  $F(\Phi) \equiv \xi\Phi^2$ , central to all metric variations and junction conditions. Matter fields (fermions  $\psi$ ) couple through Yukawa interactions  $\mathcal{L}_Y = -g\Phi\bar{\psi}\psi$ , with  $m_\psi = g|\Phi|$ . In this paper, “common origin” refers to the fact that all relevant mass scales—the Planck mass  $M_{\text{Pl}}$ , the dark-matter fermion mass  $m_\psi = g\Phi_0$ , and the overall scale of  $V_J(\Phi)$ —are set by a single vacuum expectation value  $\langle\Phi\rangle = \Phi_0$ ; the dark matter is a fermionic partner whose stability is guaranteed by the same UV gauge structure (the discrete  $\mathbb{Z}_2^\psi$  symmetry). We do not claim a microphysical derivation of the dark-energy scale  $V_c$ , which is calibrated empirically.

## 2 Theoretical foundation

### 2.1 Action and field equations

We adopt the  $(-, +, +, +)$  metric signature and the Landau–Lifshitz curvature conventions ( $R^\rho_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \dots$ ,  $R_{\mu\nu} = R^\rho_{\mu\rho\nu}$ ,  $R = g^{\mu\nu}R_{\mu\nu}$ ).

The Jordan-frame action is

$$S = \int d^4x \sqrt{-g} \left[ \frac{1}{2} \xi \Phi^2 R - \frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - V(\Phi) \right] + S_{\text{matter}}, \quad (1)$$

with the symmetry-breaking potential

$$V(\Phi) = \frac{\lambda_0}{4} (\Phi^2 - \Phi_0^2)^2 + V_c, \quad \Phi_0 = \frac{M_{\text{Pl}}}{\sqrt{\xi}}, \quad (2)$$

where  $V_c$  is a vacuum-energy constant,  $M_{\text{Pl}} \approx 2.435 \times 10^{18}$  GeV, and  $\lambda_0$  is the self-coupling. The  $\mathbb{Z}_2$  symmetry  $\Phi \rightarrow -\Phi$  produces degenerate vacua  $\pm\Phi_0$  separated by a potential barrier at  $\Phi = 0$ .

Variation with respect to  $g_{\mu\nu}$  yields the modified Einstein equation

$$FG_{\mu\nu} = T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{(\Phi)} + \nabla_\mu \nabla_\nu F - g_{\mu\nu} \square F, \quad (3)$$

with canonical scalar energy-momentum tensor  $T_{\mu\nu}^{(\Phi)} \equiv \partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} g_{\mu\nu} (\partial\Phi)^2 - g_{\mu\nu} V(\Phi)$ . The scalar equation of motion is

$$\square\Phi - V'(\Phi) + \xi\Phi R = 0. \quad (4)$$

We distinguish three field variables: the Jordan-frame field  $\Phi$ ; the logarithmic parameter  $\varphi \equiv \ln(\Phi/\Phi_0)$  with conformal factor  $\Omega^2 = e^{2\varphi}$ ; and the Einstein-frame canonical field  $\chi$  ( $d\chi/d\varphi = M_{\text{Pl}}\sqrt{6+1/\xi}$ ).

## 2.2 $\mathbb{Z}_2$ symmetry and double vacua

The action is invariant under  $\Phi \rightarrow -\Phi$ ; extrema satisfy  $\Phi \in \{0, \pm\Phi_0\}$ . The domain-wall tension is

$$\sigma_{\text{wall}} = \int_{-\Phi_0}^{+\Phi_0} d\Phi \sqrt{2[V(\Phi) - V(\Phi_0)]} = \frac{4}{3} \sqrt{\frac{\lambda_0}{2}} \Phi_0^3 > 0. \quad (5)$$

Cosmological domain-wall networks are avoided through initial-condition locking during inflation and inflationary dilution: since  $m_\Phi = \sqrt{2\lambda_0}\Phi_0 \simeq 2.7 \times 10^{14}$  GeV  $\gg H_{\text{inf}} \simeq 1.64 \times 10^{13}$  GeV, the field relaxes to one vacuum branch ( $+\Phi_0$  or  $-\Phi_0$ ) within a few  $m_\Phi^{-1}$ , well before the CMB scales exit the horizon; inflation then stretches this choice beyond the present horizon, so the standard post-inflationary Kibble wall-formation argument does not apply. The relevant suppression is in *number density*, not energy density: the number of walls per comoving volume is diluted by  $n_{\text{wall}} \propto a^{-3} \sim e^{-3N}$ , giving a probability  $\sim e^{-3N} \sim 10^{-65}$  that a single wall intersects our observable patch. (For completeness: the wall tension  $\sigma_{\text{wall}} \sim (4/3)\sqrt{\lambda_0/2}\Phi_0^3 \sim 10^{50}$  GeV<sup>3</sup> vastly exceeds the CMB bound  $\sigma \leq (1 \text{ MeV})^3$ , so any wall in our patch would be catastrophic—but the number-density suppression by inflation makes this exponentially unlikely.) Bias-potential elimination remains an open problem.

## 2.3 Unified mass scales from the scalar VEV

The key unification mechanism is that all mass scales originate from the VEV of  $\Phi$ :

### *a. Planck mass.*

The effective gravitational constant is

$$G_{\text{eff}} = \frac{1}{8\pi\xi\Phi^2} \implies G_N = \frac{1}{8\pi\xi\Phi_0^2}. \quad (6)$$

At  $\Phi = \Phi_0$ ,  $G_{\text{eff}} = G_N$  matches Newton's constant. The active order parameter is  $\mathcal{A} \equiv (\Phi/\Phi_0)^2$ , with  $G_{\text{eff}} = G_N/\mathcal{A}$ .

**On the relation**  $M_{\text{Pl}}^2 = \xi \Phi_0^2$ .

In induced gravity this is a definition, not a tuning: the Planck mass is not an independent input but is generated by the VEV,  $M_{\text{Pl}}^2 \equiv \xi \langle \Phi \rangle^2$ . The potential  $V(\Phi) = \frac{\lambda_0}{4}(\Phi^2 - \Phi_0^2)^2 + V_c$  has its extrema at  $\Phi = \pm \Phi_0$  exactly (the constant  $V_c$  does not shift the minimum), so the VEV coincides with  $\Phi_0$  as a consequence of the potential's construction. A quantum shift of the VEV would rescale  $M_{\text{Pl}}$  proportionally, leaving the dimensionless identity  $\Omega = \Phi/\Phi_0$  (Eq. (10))—which depends only on the coupling form  $F = \xi \Phi^2$ , not on the numerical value of  $\Phi_0$ —intact. The conformal decoupling derived in Sec. II E is therefore a structural property of the induced-gravity coupling, not a fine-tuned coincidence.

### ***b. Fermion masses.***

Fermions  $\psi$  couple through the Yukawa interaction  $\mathcal{L}_Y = -g\Phi\bar{\psi}\psi$ , giving  $m_\psi = g|\Phi|$ . At the minimum,  $m_\psi^{(0)} = g\Phi_0$ . A full Standard Model embedding (including Higgs-mediated masses and RG stability) is constructed in Sec. VIII.

### ***c. Weak scale (optional Higgs portal).***

If a Higgs portal  $\lambda_{\Phi H}\Phi^2|H|^2$  is included, the Higgs VEV becomes

$$v_H^2 = \frac{\mu_H^2 - \lambda_{\Phi H}\Phi_0^2}{2\lambda_H}, \quad (7)$$

linking the electroweak scale to  $\Phi_0$  and hence to  $M_{\text{Pl}}$ . This preserves the  $\mathbb{Z}_2$  symmetry ( $\Phi^2$  is even) and is technically natural [11].

## **2.4 Conformal transformation to Einstein frame**

Under  $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu} = e^{2\varphi} g_{\mu\nu}$  with  $\Omega^2 \equiv \xi \Phi^2/M_{\text{Pl}}^2$ , the exact Einstein-frame potential is

$$V_E(\varphi) = \frac{\lambda_0 M_{\text{Pl}}^4}{4\xi^2} (1 - e^{-2\varphi})^2 + V_c e^{-4\varphi}. \quad (8)$$

Two limits:  $\varphi \gg 1$  gives  $V_E \rightarrow \lambda_0 M_{\text{Pl}}^4/(4\xi^2) \equiv V_0$  (inflationary plateau);  $\varphi \rightarrow 0$  gives  $V_E \rightarrow V_c$  (frozen cosmological constant).

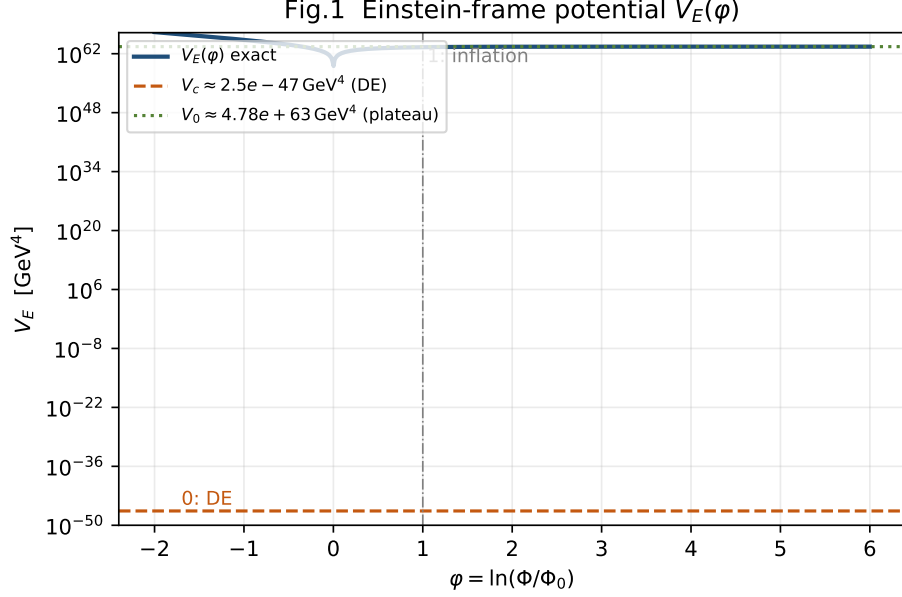
With the canonical field  $\chi$ , the potential assumes the Starobinsky form [4]

$$\frac{d\chi}{d\varphi} = M_{\text{Pl}} \sqrt{6 + \frac{1}{\xi}}, \quad V_E(\chi) = \frac{\lambda_0 M_{\text{Pl}}^4}{4\xi^2} \left(1 - e^{-\beta\chi/M_{\text{Pl}}}\right)^2, \quad \beta \equiv \frac{2}{\sqrt{6 + 1/\xi}}, \quad (9)$$

where  $\beta \simeq 0.810$  for  $\xi \sim 11.1$ . The  $V_c$  term is negligible on the plateau but determines the late-time dark-energy density.

## **2.5 Conformal decoupling and automatic fifth-force screening**

A single exact identity of this model decouples the scalar field from the dark-matter fermion  $\psi$ , while the scalar's large mass screens the fifth force for Standard Model



**Fig. 1** Exact Einstein-frame potential  $V_E(\varphi)$  for locked  $N = 50$ ,  $\xi = 11.1$  ( $\lambda_0 \simeq 6.70 \times 10^{-8}$  from exact slow-roll  $A_s$ ). The flat plateau at  $\varphi \gg 1$  ( $\sim V_0$ ) drives slow-roll inflation; the minimum at  $\varphi \simeq 0$  is set by the calibrated  $V_c$ . The hierarchy  $V_0/V_c \sim 10^{110}$  is spanned by the conformal factor  $e^{-4\varphi}$  but not dynamically explained. Regenerated by `scripts/fig1.einstein-potential.py`.

matter. Because  $F(\Phi) = \xi\Phi^2$ , the conformal factor is exactly proportional to the field:

$$\Omega = \frac{\sqrt{\xi}\Phi}{M_{\text{Pl}}} = \frac{\Phi}{\Phi_0} \implies \Phi = \Phi_0 \Omega, \quad (\text{exact, all } \Phi). \quad (10)$$

This proportionality is specific to the induced-gravity coupling; a generic  $F(\Phi)$  would not enjoy it.

We distinguish two classes of fermions. For fermions whose mass arises entirely from the direct Yukawa coupling  $-g\Phi\bar{\psi}\psi$ , the conformal identity produces exact decoupling. Under  $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$  and  $\tilde{\psi} = \Omega^{-3/2}\psi$  (equivalently  $\psi = \Omega^{3/2}\tilde{\psi}$ ), the Yukawa term transforms as

$$-g\Phi\bar{\psi}\psi\sqrt{-g} \longrightarrow -g\Phi_0\bar{\tilde{\psi}}\tilde{\psi}\sqrt{-\tilde{g}} : \quad (11)$$

the  $\Omega$  factors cancel exactly ( $\Omega \cdot \Omega^3 \cdot \Omega^{-4} = 1$ ), giving an *exact*,  $\chi$ -independent Einstein-frame fermion mass  $m_E = g\Phi_0$ . The tree-level  $\chi\bar{\psi}\tilde{\psi}$  coupling therefore vanishes identically, so the PPN parameter  $\gamma = 1$  holds exactly within the  $\psi$  sector. At loop level, graviton exchange can induce an effective  $\chi\psi\tilde{\psi}$  vertex of parametric size  $\leq m_\psi m_\chi / M_{\text{Pl}}^2 \sim 10^{-11}g$ —utterly negligible for reheating, dark matter production, and fifth-force tests, so the tree-level vanishing is the operative feature. This  $\psi$ -sector decoupling is observationally inert for solar-system tests, which probe Standard Model matter (no  $\psi$  sources are present in the solar system); it is instead decisive for the reheating and dark-matter analysis (Secs. IV–V).

Standard-model fermions obtain their mass through the Higgs mechanism, so the conformal identity does *not* cancel their  $\chi$  coupling; a residual coupling  $\sim m_f/M_{\text{Pl}}$  remains (treated separately in Sec. VIII). The solar-system fifth-force bound is satisfied by the dominant screening mechanism, namely the scalar’s large canonical mass near the minimum,

$$m_\chi^2 = \frac{V_E''(0)}{M_{\text{Pl}}^2(6 + 1/\xi)} \approx \frac{2\lambda_0 M_{\text{Pl}}^2}{\xi^2(6 + 1/\xi)} \approx (3.25 \times 10^{13} \text{ GeV})^2, \quad (12)$$

(at locked  $N = 50$ ,  $\xi = 11.1$ ;  $3.19 \times 10^{13} \text{ GeV}$  at  $N = 51$ ), which gives a Yukawa range  $m_\chi^{-1} \sim 10^{-27} \text{ cm}$ , exponentially suppressing any residual interaction at macroscopic distances and satisfying the Cassini bound  $|\gamma - 1| < 2.3 \times 10^{-5}$  [12]. This short-range suppression is the same mechanism operative in any heavy-scalar-tensor theory; it is distinct from a chameleon field-environment mechanism, which is not invoked here.

***Key differences from standard Higgs inflation.***

While the plateau geometry is shared: (a)  $\xi \sim 11.1$  is two orders below  $\xi \sim 10^4$ , yielding a different unitarity regime; (b)  $M_{\text{Pl}}$  is dynamically generated via induced gravity; (c) the same  $V_J(\Phi)$  supplies both the inflationary plateau and the late-time vacuum constant  $V_c$ ; (d)  $\Phi$  is a gauge singlet, so the large top-Yukawa term that drives the Higgs  $\beta_\xi$  is absent.

## 2.6 EFT cutoff scale

The status of the cutoff is frame-dependent and, for non-minimal scalar-tensor inflation, *contested*. From the Einstein-frame EFT perspective [3], the action is standard GR plus a canonical scalar; higher operators are suppressed by  $M_{\text{Pl}}$  and  $H_{\text{inf}} \ll M_{\text{Pl}}$  ensures self-consistency. From the Jordan-frame perspective [13, 14], the cutoff is  $\Lambda_J \sim M_{\text{Pl}}/\xi \sim 2.2 \times 10^{17} \text{ GeV}$ , and the CMB pivot value  $\Phi_{\text{CMB}}$  exceeds it by  $\sim 30$ . Whether the low Jordan-frame cutoff persists as a physical limitation of the Einstein-frame EFT is an open question without community consensus [15]. We adopt the Einstein-frame EFT perspective for computing slow-roll observables (where they are robust) and flag the Jordan-frame cutoff tension as an open issue (Discussion #8).

## 3 Inflation: Starobinsky-type slow-roll plateau

***Conditional statement.***

The derivation describes the geometric fact that a slow-roll plateau exists whenever  $\chi \gg M_{\text{Pl}}$ . The naturalness of this initial condition is not endogenously explained (Discussion #1).

### 3.1 Energy scale and CMB normalization

**Definition of  $N$  and CMB normalization.**

Throughout this paper the  $e$ -fold number is the cosmological quantity

$$N \equiv \ln\left(\frac{a_{\text{end}}}{a_*}\right), \quad k_* = a_* H_*, \quad (13)$$

with pivot  $k_* = 0.05 \text{ Mpc}^{-1}$ . Post-inflationary matching assumes matter-like ( $w = 0$ ) condensate domination until  $T_{\text{reh}}$ ,

$$a_{\text{end}} = a_{\text{reh}} \left(\frac{\rho_{\text{reh}}}{\rho_{\text{end}}}\right)^{1/3}, \quad a_{\text{reh}} = a_{\text{eq}} \left(\frac{\rho_{\text{eq}}}{\rho_{\text{reh}}}\right)^{1/4}, \quad a_{\text{eq}} = \frac{\Omega_r}{\Omega_m}, \quad (14)$$

with  $\Omega_m \simeq 0.315$ ,  $\Omega_r \simeq 9 \times 10^{-5}$ ,  $g_* = 106.75$ . At a given  $N$ , the plateau variable  $x_*$  is fixed by the exact potential-slow-roll integral

$$N = \frac{e^{x_*} - x_* - (e^{x_{\text{end}}} - x_{\text{end}})}{2\beta^2}, \quad e^{-x_{\text{end}}} = \frac{1}{1 + \sqrt{2}\beta}, \quad \beta \equiv \frac{2}{\sqrt{6 + 1/\xi}}, \quad (15)$$

and observables are evaluated at that  $x_*$ :

$$r = 16\varepsilon_*, \quad n_s = 1 - 6\varepsilon_* + 2\eta_*, \quad A_s = \frac{V_0(1 - e^{-x_*})^2}{24\pi^2 M_{\text{Pl}}^4 \varepsilon_*}, \quad (16)$$

with

$$\varepsilon = \frac{2\beta^2 e^{-2x}}{(1 - e^{-x})^2}, \quad \eta = \frac{2\beta^2 e^{-x}(2e^{-x} - 1)}{(1 - e^{-x})^2}, \quad V_0 = \frac{\lambda_0 M_{\text{Pl}}^4}{4\xi^2}. \quad (17)$$

Solving  $A_s = 2.100 \times 10^{-9}$  at fixed  $(N, \xi)$  determines  $\lambda_0$  uniquely (linear in  $\lambda_0$  at fixed  $x_*$ ). Attractor-style closed forms  $r \simeq 8/(\beta^2 N^2)$ ,  $n_s \simeq 1 - 2/N$  remain useful approximations but *must not* be mixed with this  $N$  definition: at  $N = 50$  the exact potential slow-roll gives  $r \approx 0.00425$ , whereas the closed form yields  $r \approx 0.00487$ .

**Fiducial point.**

For  $\xi = 11.1$  and  $N = 50$  under Eq. (13):

$$\lambda_0 \simeq 6.70 \times 10^{-8}, \quad r \simeq 0.00425, \quad n_s \simeq 0.9616, \quad (18)$$

$$H_{\text{inf}} = \sqrt{\frac{V_0}{3M_{\text{Pl}}^2}} \simeq 1.64 \times 10^{13} \text{ GeV}, \quad U^{1/4} \simeq 8.3 \times 10^{15} \text{ GeV}, \quad m_\chi \simeq 3.25 \times 10^{13} \text{ GeV}, \quad (19)$$

where  $m_\chi^2 = 8V_0/(M_{\text{Pl}}^2 \beta_o^2) = 2\lambda_0 M_{\text{Pl}}^2/[\xi^2(6 + 1/\xi)]$  with  $\beta_o = \sqrt{6 + 1/\xi}$ . The conformal-anomaly reheating estimate  $T_{\text{reh}} \sim 10^9 \text{ GeV}$  maps through Eqs. (14)–(15) to  $N \simeq 51$  (not 50); the corresponding self-consistent temperature for  $N = 50$  is  $T_{\text{reh}} \sim 4 \times 10^7 \text{ GeV}$ . Matching assumptions ( $k_*$ ,  $w$  during reheating,  $\Omega$ 's) shift these temperatures by  $\mathcal{O}(1)$ – $\mathcal{O}(10)$ .

### ***Sensitivity table.***

Table 1 lists exact slow-roll predictions at  $\xi = 11.1$  under the locked  $N$  definition. Planck  $2\sigma$  ( $n_s \in [0.9565, 0.9733]$ ) selects  $N \approx 45\text{--}56$  with  $r \in [0.0034, 0.0052]$ ; the tighter  $1\sigma$  lower edge  $n_s \geq 0.9607$  selects  $N \geq 49$ . The upper edge is not set by  $n_s$  but by reheating: the tabulated  $T_{\text{reh}}^*(N)$  grows as  $e^{3.05N}$ , so  $T_{\text{reh}}^*(56) \simeq 3.8 \times 10^{15}$  GeV already exceeds the instantaneous-reheating ceiling  $T_{\text{reh}}^{\text{inst}} = (30\rho_{\text{end}}/\pi^2 g_*)^{1/4} \simeq 2.6 \times 10^{15}$  GeV obtained at  $N = 56$ ; hence  $N_{\text{max}} \simeq 55.9$  and  $N = 57, 58$  are physically excluded (Appendix D).

### ***Two-sided physical window.***

The admissible band is bounded on *both* sides by physics, not by the  $n_s$  likelihood alone. On the high side,  $T_{\text{reh}}^*(N) \propto e^{3N}$  while the instantaneous-reheating ceiling falls only as  $N^{-1/2}$ , giving  $N_{\text{max}} \simeq 55.9$  as shown above. On the low side, the model cannot be pushed to small  $N$ : the tensor-to-scalar ratio falls to  $r = 0.01$  at  $N_* \simeq 32$  in the locked- $N$  convention (the attractor closed form  $r \simeq 8/(\beta^2 N^2)$  gives  $N_* \simeq 35$ ), so  $r > 0.01$  requires  $N \gtrsim 32$ , for which the matching condition (14) returns  $T_{\text{reh}} \simeq 5.9 \times 10^{-17}$  GeV—fourteen orders of magnitude below the BBN floor of 10 MeV—so such a point is physically impossible for any reheating mechanism. The viable interval is therefore  $N \in [45, 56]$ , and Planck  $2\sigma$  on  $n_s$  happens to select almost exactly that range. This is what makes the model falsifiable in a definite way: a LiteBIRD-class detection  $r > 0.01$  would demand  $N \lesssim 32$  and hence a reheating temperature incompatible with BBN, so it would exclude the model class outright rather than merely disfavour it.

### ***Reheating equation of state and the $N$ – $T_{\text{reh}}$ mapping.***

The mapping above assumes the condensate behaves as pressureless dust ( $w = 0$ ) throughout the post-inflationary phase. This is an approximation: the potential is not exactly harmonic at the largest amplitudes (Appendix D), and the exact integration of Sec. D0 gives  $\langle w \rangle \simeq -0.02$  averaged over the first few  $e$ -folds, approaching zero thereafter. A constant- $w$  reheating epoch with  $\bar{w}_{\text{eff}}$  shifts the  $e$ -fold count by  $\Delta N = \frac{1-3\bar{w}_{\text{eff}}}{12(1+\bar{w}_{\text{eff}})} \ln(T_{\text{reh}}/T_{\text{end}})$ ; for  $\bar{w}_{\text{eff}} \in [0, 1/3]$  this gives  $|\Delta N| \lesssim 1$ , so the mapping is stable at the stated  $\mathcal{O}(1)$  level and the quoted band is unaffected.

### ***Quantum diffusion.***

During slow-roll, quantum fluctuations cause the field to undergo a random walk with variance  $\langle \delta\chi^2 \rangle \simeq H_{\text{inf}}^3 t / (4\pi^2)$  per Hubble time. Over one  $e$ -fold  $\Delta t = H_{\text{inf}}^{-1}$ , the accumulated quantum jitter is  $\Delta\chi_{\text{quant}} \sim H_{\text{inf}}/(2\pi)$ , while the classical drift is  $\Delta\chi_{\text{class}} \sim \dot{\chi}/H_{\text{inf}} \sim M_{\text{Pl}}\sqrt{2\epsilon}$ . The ratio  $\Delta\chi_{\text{quant}}/\Delta\chi_{\text{class}} \sim H_{\text{inf}}/(2\pi M_{\text{Pl}}\sqrt{2\epsilon}) \sim 6 \times 10^{-6}/\sqrt{\epsilon}$  is utterly negligible except at the very end of inflation ( $\epsilon \rightarrow 1$ ), where it reaches  $\sim 6 \times 10^{-6}$ —still five orders below unity. Stochastic effects therefore do not invalidate the classical slow-roll trajectory for this model.

### ***Attractor forms and non-Gaussianity.***

Large-field attractor expressions  $n_s \simeq 1 - 2/N$ ,  $r \simeq 8/(\beta^2 N^2)$ ,  $n_t = -r/8$  are useful analytic guides; numerical values quoted in Table 1 use the exact integral (15) and

**Table 1** Exact potential slow-roll at  $\xi = 11.1$  under the locked definition  $N = \ln(a_{\text{end}}/a_*)$  (Eqs. (13)–(16)).  $\lambda_0$  is fixed by  $A_s = 2.100 \times 10^{-9}$ .  $T_{\text{reh}}^*$  is the temperature for which cosmological matching (14) returns  $N_{\text{derived}} = N$ . Planck  $2\sigma$  allows  $N \approx 45\text{--}56$  ( $r \leq 0.0052$ ).

| $N$ | $\lambda_0$           | $n_s$  | $r$     | $H_{\text{inf}}$ [GeV] | $T_{\text{reh}}^*$ [GeV] |
|-----|-----------------------|--------|---------|------------------------|--------------------------|
| 48  | $7.24 \times 10^{-8}$ | 0.9600 | 0.00459 | $1.70 \times 10^{13}$  | $1.0 \times 10^5$        |
| 49  | $6.96 \times 10^{-8}$ | 0.9608 | 0.00441 | $1.67 \times 10^{13}$  | $2.1 \times 10^6$        |
| 50  | $6.70 \times 10^{-8}$ | 0.9616 | 0.00425 | $1.64 \times 10^{13}$  | $4.4 \times 10^7$        |
| 51  | $6.45 \times 10^{-8}$ | 0.9623 | 0.00409 | $1.61 \times 10^{13}$  | $9.0 \times 10^8$        |
| 52  | $6.21 \times 10^{-8}$ | 0.9630 | 0.00394 | $1.58 \times 10^{13}$  | $1.8 \times 10^{10}$     |
| 55  | $5.58 \times 10^{-8}$ | 0.9650 | 0.00355 | $1.50 \times 10^{13}$  | $1.6 \times 10^{14}$     |

potential slow roll (16) at the locked  $N$ . Being of the Starobinsky type, the model also predicts negligible primordial non-Gaussianity:  $f_{\text{NL}}^{\text{local}} = -\frac{5}{12}(1 - n_s) \sim -0.02$ , with equilateral and orthogonal shapes suppressed by the slow-roll parameters to  $\mathcal{O}(10^{-2})$ , all well within Planck 2018 constraints [16] ( $f_{\text{NL}}^{\text{local}} = -0.9 \pm 5.1$ ).

### 3.2 Slow-roll parameters and observables

In the  $\chi \gg M_{\text{Pl}}$  limit, the potential slow-roll parameters are

$$\epsilon_V \equiv \frac{M_{\text{Pl}}^2}{2} \left( \frac{V'_E}{V_E} \right)^2 \simeq 2\beta^2 e^{-2\beta\chi/M_{\text{Pl}}} \ll 1, \quad (20)$$

$$\eta_V \equiv M_{\text{Pl}}^2 \frac{V''_E}{V_E} \simeq -2\beta^2 e^{-\beta\chi/M_{\text{Pl}}}, \quad (21)$$

$$N \simeq \frac{e^{\beta\chi_N/M_{\text{Pl}}}}{2\beta^2}. \quad (22)$$

We use the potential slow-roll (PSR) formulation throughout; the Hubble slow-roll (HSR) parameters  $\epsilon_H \equiv -\dot{H}/H^2$  and  $\eta_H \equiv \ddot{\chi}/(H\dot{\chi})$  coincide with  $\epsilon_V$  and  $\eta_V$  to leading order in slow-roll, with relative differences  $\Delta\epsilon/\epsilon \sim \mathcal{O}(\epsilon, \eta) \leq 10^{-2}$  in the deep slow-roll regime and rising to  $\sim \mathcal{O}(1)$  only at the very end of inflation ( $\epsilon \rightarrow 1$ ). The  $e$ -fold number  $N$ —the quantity that enters the CMB observables—is insensitive to the PSR/HSR distinction at the  $\leq 0.1$  level, well within the  $\mathcal{O}(1)$  systematic uncertainty already quoted for the post-inflationary phase.

#### *Cutoff disclaimer.*

The slow-roll expressions are leading-order results in the Einstein-frame EFT ( $\Lambda_E \sim M_{\text{Pl}}$ ). Higher-derivative corrections enter at order  $(H_{\text{inf}}/M_{\text{Pl}})^2 \sim 3 \times 10^{-11}$ ; the functional forms  $n_s(N)$  and  $r(N)$  are insensitive to the cutoff.

#### *Perturbation stability.*

In the Einstein frame the scalar field  $\chi$  is canonically normalized with a standard kinetic term  $\frac{1}{2}(\partial\chi)^2$ , so the scalar sound speed is  $c_s^2 = 1$  identically—no gradient

instability can arise. The tensor sector is standard GR with  $c_T = 1$  (see also Discussion #16). The comoving curvature perturbation  $\zeta$  is conserved on superhorizon scales for adiabatic single-field models [19]; no entropy/isocurvature modes are sourced because the single field  $\Phi$  drives both the background and the perturbations. The standard single-field slow-roll formulas for  $n_s$ ,  $r$ , and the scalar power spectrum therefore apply without modification.

The scalar spectral index with NLO correction is

$$n_s \simeq 1 - \frac{2}{N} - \frac{3}{2N^2} + \mathcal{O}(N^{-3}). \quad (23)$$

(The NLO coefficient  $-\frac{3}{2}$  applies in the Starobinsky limit  $\beta = \sqrt{2/3}$ ; for general  $\beta$  it acquires a  $\beta$ -dependence. **All tabulated  $n_s$  and  $r$  in this paper use the exact potential slow-roll evaluation (16) at the locked  $N$ , not this NLO closed form.** A quantitative slow-roll scan shows the NLO closed form differs from exact PS by  $\leq 3 \times 10^{-4}$  in  $n_s$  at  $\xi = 11.1$  ( $\beta = 0.810$ ) and throughout  $\xi \geq 5$ —well below Planck  $\sigma(n_s) = 0.0042$ —but the shift grows toward  $\xi \rightarrow 1$ ; precision comparisons at small  $\xi$  must use the exact integral (15).)

For the locked definition, exact potential slow roll at  $\xi = 11.1$  gives  $n_s \simeq 0.9616$  at  $N = 50$  ( $-0.79\sigma$  from Planck) and  $n_s \simeq 0.9650$  at  $N = 55$  ( $+0.02\sigma$ ); see Table 1. Planck 2018 [16]:  $0.9649 \pm 0.0042$  ( $1\sigma$ ). The prediction approaches the Planck central value as  $N$  increases; the  $2\sigma$  band corresponds to  $N \approx 45$ – $56$  in this convention.

The tensor-to-scalar ratio under the same convention is  $r \simeq 0.00425$  at  $N = 50$  and  $r \simeq 0.00355$  at  $N = 55$ ; across Planck  $2\sigma$   $N \approx 45$ – $56$ ,  $r \in [0.0034, 0.0052]$ . Planck+BICEP/Keck [16, 17]:  $r < 0.036$  (95% CL).

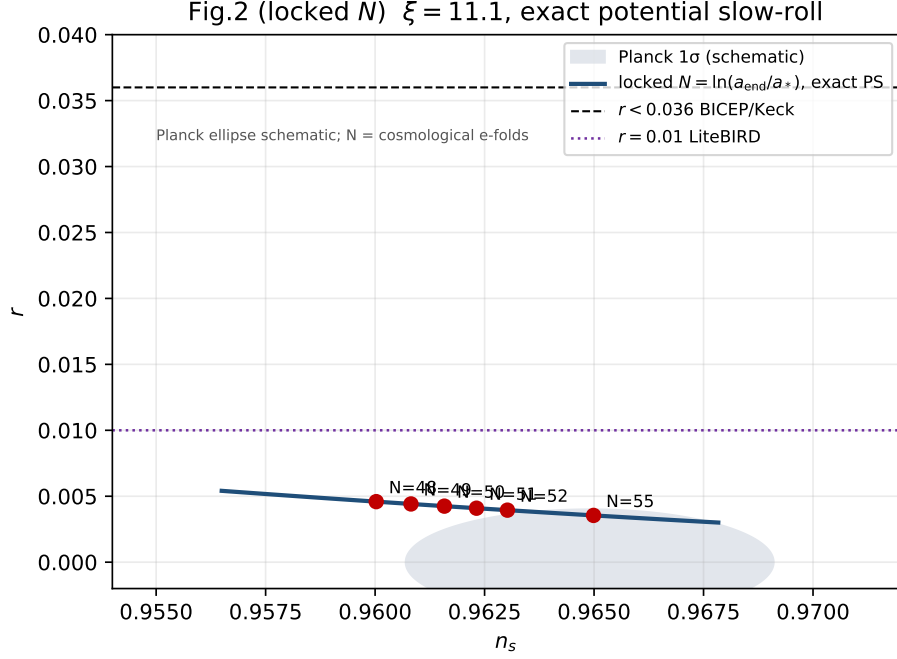
A measurement  $r > 0.01$  at 95% CL by LiteBIRD [18] would rule out the entire physically allowed model class in this convention: exact slow-roll inversion puts  $r = 0.01$  at  $N \approx 32$  (for  $\xi = 11.1$ ), outside the Planck-compatible band, and reheating temperatures required for such small  $N$  fall far below BBN. Throughout the  $2\sigma$  window,  $r \leq 0.0052$  at  $\xi = 11.1$  (only weakly  $\xi$ -dependent once  $x_*(N)$  is fixed by the exact integral).

Figure 2 shows the model predictions superimposed on Planck+BICEP/Keck-style constraints.

## 4 End of slow-roll and reheating

### 4.1 End of slow-roll and coherent oscillations

Slow-roll ends at  $\epsilon = 1$ . The exact  $\epsilon$  derived from  $V_E = V_0(1 - e^{-\beta\chi/M_{\text{Pl}}})^2$  is  $\epsilon_{\text{exact}} = 2\beta^2 e^{-2x}/(1 - e^{-x})^2$  with  $x \equiv \beta\chi/M_{\text{Pl}}$ ; the large-field limit  $(1 - e^{-x}) \approx 1$  reduces it to



**Fig. 2**  $n_s$ - $r$  under the locked definition  $N = \ln(a_{\text{end}}/a_*)$  at  $\xi = 11.1$  (exact potential slow-roll,  $\lambda_0$  from  $A_s$ ; Table 1). Markers: locked- $N$  points. Planck ellipse schematic; BICEP/Keck  $r < 0.036$ ; LiteBIRD line  $r = 0.01$ . Regenerated by `scripts/fig2_ns_r.py` (locked- $N$  convention; same as `lock_n_convention.py`).

Eq. (22). Solving  $\epsilon_{\text{exact}} = 1$  with  $u \equiv e^{-x}$  gives  $u = 1/(1 + \sqrt{2}\beta) \approx 0.466$ , so

$$e^{-\beta\chi_{\text{end}}/M_{\text{Pl}}} = \frac{1}{1 + \sqrt{2}\beta} \approx 0.466,$$

$$V_{\text{end}} = V_0 \left(1 - e^{-\beta\chi_{\text{end}}/M_{\text{Pl}}}\right)^2 \approx 0.285 V_0 \approx 1.36 \times 10^{63} \text{ GeV}^4 \quad (24)$$

( $V_0 \simeq 4.78 \times 10^{63} \text{ GeV}^4$  at locked  $N = 50$ ).

Thereafter the field oscillates around  $\Phi_0$  with mass  $m_\Phi = \sqrt{2\lambda_0}\Phi_0 \approx 2.7 \times 10^{14} \text{ GeV}$ . Near the minimum,  $V_J(\Phi) \approx \lambda_0\Phi_0^2(\Phi - \Phi_0)^2 + \mathcal{O}((\Phi - \Phi_0)^3)$ , i.e. the potential is quadratic to leading order; the quartic  $(\Phi^2 - \Phi_0^2)^2$  form produces a radiation-like equation of state ( $w \simeq 1/3$ ) only for oscillations around  $\Phi = 0$ , which is never reached since the field is confined to one vacuum branch ( $+\Phi_0$  or  $-\Phi_0$ ) by the initial-condition locking of Sec. II. Consequently the amplitude redshifts as  $A(t) \propto a^{-3/2}$  and the condensate evolves as pressureless dust ( $w = 0$ ). This last statement is an approximation: the post-inflationary excursion reaches  $|x| \equiv |\beta\chi/M_{\text{Pl}}| \simeq 0.88$  (Sec. III and App. D), where the cubic and quartic corrections are  $\mathcal{O}(1)$ , and the exact integration gives  $\langle w \rangle \simeq -0.02$  averaged over the first  $e$ -folds rather than exactly zero. The  $a^{-3}$  scaling,

the  $N$ – $T_{\text{reh}}$  mapping and the dark-energy budget are therefore accurate at the few-per-cent level, well inside the  $\mathcal{O}(1)$  matching uncertainty quoted in Sec. III. We adopt the standard approximation that the transition from slow-roll to coherent oscillations is instantaneous; a more detailed treatment of the transient post-inflationary phase (wherein the field may traverse the steep region of the potential before settling into harmonic motion) would shift the numerical mapping between  $N$  and  $T_{\text{reh}}$  by an  $\mathcal{O}(1)$  factor in the expansion history and is deferred to a dedicated lattice study.

#### *Overshoot across the barrier.*

At the end of inflation the kinetic energy of the field is  $K_{\text{end}} = \frac{1}{2}\dot{\chi}_{\text{end}}^2 = \frac{\epsilon_H}{3-\epsilon_H}V_{\text{end}}$ , where  $\epsilon_H \equiv -\dot{H}/H^2$ ; the slow-roll relation  $K = \epsilon V$  is only valid to leading order in  $\epsilon$ , and at  $\epsilon_V = 1$  the exact attractor gives  $\epsilon_H \simeq 0.499$ , hence  $K_{\text{end}} \simeq 0.199 V_{\text{end}} \simeq 2.7 \times 10^{62} \text{ GeV}^4$  (obtained by numerically integrating the exact Klein–Gordon equation from the slow-roll regime to  $\epsilon_V = 1$ ; see the discussion of numerical validation in Appendix D). The Jordan-frame barrier height at  $\Phi = 0$  is  $\Delta V_J = V_J(0) - V_J(\Phi_0) = \lambda_0 \Phi_0^4/4 \equiv V_0 \simeq 4.78 \times 10^{63} \text{ GeV}^4$ . Although the Jordan-frame barrier at  $\Phi = 0$  has height  $\Delta V_J = V_0$ , that is *not* the quantity governing the dynamics: the propagating degree of freedom is the canonical field  $\chi$  in the Einstein frame, whose potential is  $V_E(x) = V_0(1 - e^{-x})^2$  with  $x \equiv \beta\chi/M_{\text{Pl}}$ , and the conformal factor pushes the  $\Phi \rightarrow 0$  limit to  $V_E \rightarrow \infty$ . The correct criterion is therefore whether the conserved total energy density suffices to reach  $\Phi = 0$ , which would require infinite energy. The field is bounded instead by the turning points of  $V_E(x) = \rho_{\text{end}}$ , where  $\rho_{\text{end}} = K_{\text{end}} + V_{\text{end}} = 1.199 V_{\text{end}} = 0.342 V_0$ ; solving  $(1 - e^{-x})^2 = 0.342$  gives  $x_{\text{turn}} = +0.879$  and  $-0.461$ , i.e.  $\Phi/\Phi_0 = e^{x/2} \in [0.79, 1.55]$ . The field therefore oscillates between  $\Phi \simeq 0.79 \Phi_0$  and  $\Phi \simeq 1.55 \Phi_0$  and never approaches  $\Phi = 0$ : no vacuum-branch crossing occurs. (The ratio  $K_{\text{end}}/\Delta V_J \simeq 0.057$  is sometimes quoted here, but it is *not* the relevant criterion, since it compares a kinetic energy with a potential difference evaluated in a different frame; the Jordan-frame barrier height  $\Delta V_J$  is likewise not the dynamical barrier. The barrier is the full height  $V_0$  between the two vacua, not the  $2.2 \times 10^{54} \text{ GeV}^4$  quoted in earlier versions.) The field displacement from  $\Phi_{\text{end}}$  to  $\Phi = 0$  ( $\varphi \rightarrow -\infty$ ) would require traversing the entire plateau, and Hubble friction during the transient post-inflationary phase removes whatever energy remains above the turning point. The initial-condition locking of Sec. II (field mass  $m_\Phi \gg H_{\text{inf}}$  during inflation) independently guarantees that the field is exponentially unlikely to be found near  $\Phi = 0$  at the end of slow-roll.

## 4.2 Vanishing of perturbative Yukawa reheating

The same conformal identity  $\Omega = \Phi/\Phi_0$  that screens the fifth force (Sec. II E) also governs reheating. The tree-level Einstein-frame  $\chi\bar{\psi}\tilde{\psi}$  coupling vanishes exactly (Eq. (10)), so there is no perturbative inflaton decay into fermions via the direct Yukawa channel—this holds for any value of  $g$ , including the value  $g \simeq 1.0 \times 10^{-7}$  selected by the dark-matter abundance in Sec. V. For  $m_\psi \gtrsim m_\chi/2$  (i.e.  $g \gtrsim \text{few} \times 10^{-5}$ ) the decay  $\chi \rightarrow \psi\bar{\psi}$  is *additionally* kinematically forbidden, so the closure is then doubly protected; at the lighter abundance-matched value the kinematic condition fails and the closure rests on the vanishing vertex alone. Because  $\Phi$  oscillates about  $\Phi_0 \neq 0$ , the

Jordan-frame fermion mass  $m_\psi = g|\Phi|$  never crosses zero; for  $g \geq 10^{-4}$  it varies adiabatically ( $\dot{m}_\psi/m_\psi^2 \sim 4.4 \times 10^{-4}/g \ll 1$ ), while for  $g \lesssim 10^{-4}$ —including the dark-matter value  $g \simeq 1.0 \times 10^{-7}$ —the variation is non-adiabatic and  $\psi\bar{\psi}$  pair production opens. This is the Jordan-frame image of the Einstein-frame gravitational production that sources the dark matter of Sec. V; like gravitational particle production, it is a classical-background/quantum-vacuum effect and does *not* use the vanishing perturbative vertex, so it does not re-open the reheating channel. The scalar self-coupling  $\lambda_{\text{self}} \sim \lambda_0\beta^4/(4\xi^2) \sim 4 \times 10^{-11}$  is too small for efficient self-resonance; a full Floquet analysis of preheating is deferred.

### 4.3 Gravitational reheating and the status of $N$

With the perturbative Yukawa and self-resonance channels absent or negligible, reheating proceeds through gravitational mechanisms [21, 23]. We distinguish the relevant channels carefully, because the status of the graviton-pair channel depends on the gravitational sector’s structure:

- (i) **Non-perturbative gravitational particle production** at the end of slow-roll: the rapidly varying metric excites SM fields from the vacuum, providing an initial radiation pulse with  $\rho_{\text{rad}}(a_{\text{end}}) \sim 2 \times 10^{-13} \rho_\chi(a_{\text{end}})$ . As shown below, this pulse dilutes *faster* than the condensate and therefore cannot reheat the universe on its own; it supplies only a seed bath. The reheating temperature is set by the continuous conformal-anomaly channel (iv).
- (ii) **Perturbative decay of the oscillating condensate to fermion pairs via graviton exchange** ( $\chi \rightarrow f\bar{f}$  through an  $s$ -channel graviton): this proceeds at  $\Gamma_{\chi \rightarrow f\bar{f}}^{(\text{grav exch})} \sim m_\chi^5/M_{\text{Pl}}^4 \sim 10^{-6} \text{ GeV}$ , slower than channel (i) by  $\sim 10^6$  and subdominant for setting  $T_{\text{reh}}$ . The direct  $\chi \rightarrow f\bar{f}$  channel through the residual  $\chi f\bar{f}$  coupling  $\sim m_f/M_{\text{Pl}}$  (Eq. (41)) is even slower,  $\sim 10^{-20} \text{ GeV}$  for the top quark.
- (iii) **Tree-level decay to graviton pairs** ( $\chi \rightarrow h\bar{h}$ ): this channel is *absent* in the present model. The reason is structural: induced gravity has the  $F(\Phi)R/2$  form with  $F = \xi\Phi^2$ , so after conformal transformation the Einstein-frame Ricci term carries a *constant* coefficient  $M_{\text{Pl}}^2/2$  with no  $\chi$  dependence; the canonical field  $\chi$  is therefore minimally coupled to the Einstein-frame gravitons (coupling only through the kinetic term), and no  $\chi h\bar{h}$  vertex is generated. Consequently  $\Gamma(\chi \rightarrow h\bar{h}) = 0$  at tree level. At the 1-loop level, graviton loops induce an effective  $\chi h\bar{h}$  vertex of size  $\Gamma_{\chi \rightarrow h\bar{h}}^{\text{eff}} \sim m_\chi^5/(16\pi^2 M_{\text{Pl}}^4) \sim 10^{-8} \text{ GeV}$ ,  $\sim 10^8$  below channel (i) and  $\sim 10^2$  below channel (ii), so this 1-loop contribution—the leading order for this structurally closed channel—is far below the dominant reheating channels and does not affect  $T_{\text{reh}}$ . This differs from  $R^2$  inflation [4], where the  $R^2$  term produces a scalaron with a  $\chi$ -dependent  $\tilde{R}$  coefficient ( $\propto e^{2\chi/\sqrt{6}M_{\text{Pl}}}$ ), opening the  $\chi \rightarrow h\bar{h}$  channel with  $\Gamma \sim m_\chi^3/(192\pi M_{\text{Pl}}^2)$ ; that result does *not* carry over to the  $F(\Phi)R/2$  theories (induced gravity or Higgs inflation), for which  $\chi \rightarrow h\bar{h}$  is closed.
- (iv) **Continuous conformal-anomaly decay**: although the gauge kinetic term is classically conformally invariant, the trace anomaly  $\langle T^\mu_\mu \rangle = (\beta_g/2g)\langle F^2 \rangle$  generates an effective  $\chi$ -gauge vertex  $\mathcal{L}_{\text{eff}} \supset (b_s\alpha_s/4\pi)(\chi/(M_{\text{Pl}}\sqrt{6+1/\xi}))G_{\mu\nu}^a G^{a\mu\nu}$

(and analogously for photons). The corresponding decay rates are  $\Gamma_{\chi \rightarrow gg} = b_3^2 \alpha_s^2 m_\chi^3 / (16\pi^3 M_{\text{Pl}}^2 (6 + 1/\xi))$  and  $\Gamma_{\chi \rightarrow \gamma\gamma} \sim \mathcal{O}(10^{-2})$  GeV, giving a total perturbative width  $\Gamma_{\text{anom}} \sim \mathcal{O}(10^{-1})$  GeV. Evaluating the QCD channel with the one-loop running coupling,  $\alpha_s(m_\chi) = 0.0262$  at  $m_\chi = 3.25 \times 10^{13}$  GeV for  $b_3 = 7$  (SM value;  $\alpha_s(m_Z) = 0.1179$ ), gives  $\Gamma_{\chi \rightarrow gg} \simeq 0.065$  GeV. Unlike channel (i), this is a genuine continuous perturbative decay, so the standard constant-rate reheating formula applies:  $T_{\text{reh}}^{(\text{anom})} = (90\Gamma_{\text{anom}}^2 M_{\text{Pl}}^2 / (\pi^2 g_*))^{1/4} \simeq 2.1 \times 10^8$  GeV. Throughout this work we retain  $T_{\text{reh}} = 10^9$  GeV as the fiducial reference temperature (it is the value entering Table 1 and the  $N$  mapping below); with the physical running coupling the anomaly channel gives  $2.1 \times 10^8$  GeV instead, which shifts the corresponding  $e$ -fold number by only  $\Delta N \simeq 0.4$  (i.e.  $N \simeq 50.6$  rather than 51), well inside the  $\mathcal{O}(1)$  matching uncertainty declared in Sec. III B. We therefore continue to quote  $N \simeq 51$  as the primary reference point.

The parametric estimate below accordingly combines channel (iv); channels (ii) and (iii) do not alter the reheating temperature at leading order. A precise prediction of  $T_{\text{reh}}$  and  $N$  cannot be obtained from a perturbative decay rate alone: the Einstein-frame  $\chi\psi\bar{\psi}$  vertex vanishes exactly, the  $\chi \rightarrow h\bar{h}$  channel is structurally closed, and the Jordan-frame Yukawa channel is kinematically forbidden. With the perturbative channels closed,  $N$  is not microphysically predicted but is constrained by the consistency requirements that the universe reheats to a temperature compatible with BBN ( $T_{\text{reh}} \geq 10$  MeV) and does not overproduce relics. The window is bounded from below by the BBN floor and from above by instantaneous reheating,  $T_{\text{reh,max}} = (30\rho_{\text{end}}/(\pi^2 g_*))^{1/4} \simeq 2.6 \times 10^{15}$  GeV at  $N = 50$  (equivalently  $T_{\text{reh,max}} \simeq 0.430 V_{\text{end}}^{1/4}$ , where  $V_{\text{end}}^{1/4} \simeq 6.1 \times 10^{15}$  GeV); this ceiling is what limits the  $n_s$ -allowed band to  $N \leq 55.9$  (Appendix D).

Using the cosmological definition (13) and matching (14) (rather than a single closed-form  $N(T_{\text{reh}})$ ), exact slow-roll inversion with  $A_s$  normalization gives the  $T_{\text{reh}}^*$  column of Table 1. Planck  $2\sigma$  on  $n_s$  then selects

$$N \approx 45\text{--}56 \quad (2\sigma \text{ Planck}), \quad \text{anomaly } T_{\text{reh}} \sim 10^9 \text{ GeV} \Rightarrow N \simeq 51, \quad (25)$$

with  $r \leq 0.0052$  throughout that band at  $\xi = 11.1$ . The self-consistent point for anomaly-dominated reheating is therefore  $N \simeq 51$  in this convention, not an a-priori fixed  $N = 50$ . Matching assumptions (pivot  $k_*$ , equation of state during reheating,  $g_*$ ) move  $N$  by  $\mathcal{O}(1)$ ; we quote the locked-definition numbers as the primary results and treat attractor closed forms as approximations only.

Earlier drafts quoted a single closed-form matching  $N = 61 + \frac{1}{4} \ln(V_{\text{end}}/M_{\text{Pl}}^4) - \frac{1}{3} \ln(V_{\text{end}}^{1/4}/T_{\text{reh}})$  (Liddle–Leach style [20]). That estimate remains a useful cross-check, but the physical window used here follows from the locked definition (13) and Planck  $n_s$ , Eq. (25).

All  $N$ -dependent numerical values in this work refer to the locked definition (13). Primary quoted points are the anomaly-reheating self-consistency value  $N \simeq 51$  and the illustrative  $N = 50$  row of Table 1; attractor closed forms  $n_s \simeq 1 - 2/N$ ,  $r \simeq 8/(\beta^2 N^2)$  are analytic guides only and are not mixed with the table.

***a. Gravitational production and conformal-anomaly decay.***

We now identify which channels can reheat the universe. After inflation, the oscillating  $\chi$  condensate dominates the energy density. Two distinct radiation sources operate:

(a) *Initial gravitational pulse* (channel (i)): Standard results [22, 23] give an initial radiation density  $\rho_{\text{rad}}(a_{\text{end}}) \sim N_{\text{eff}} H_{\text{inf}}^4 / (192\pi^2)$ , where  $N_{\text{eff}} \sim \mathcal{O}(10)$  counts SM degrees of freedom with non-conformal coupling. With  $H_{\text{inf}} \simeq 1.64 \times 10^{13}$  GeV (locked  $N = 50$ ;  $1.61 \times 10^{13}$  GeV at  $N = 51$ ), this yields  $\rho_{\text{rad}}(a_{\text{end}}) \sim 4 \times 10^{50}$  GeV<sup>4</sup>, compared to the exact condensate density  $\rho_{\text{end}} = K_{\text{end}} + V_{\text{end}} = 1.199 V_{\text{end}} \simeq 1.64 \times 10^{63}$  GeV<sup>4</sup>—a fractional radiation energy density of only  $\sim 2.3 \times 10^{-13}$ . This pulse does *not* by itself reheat the universe. During condensate domination the condensate redshifts as  $a^{-3}$  while the pulse redshifts as  $a^{-4}$ , so the radiation *fraction* decreases:

$$\frac{\rho_{\text{rad}}}{\rho_{\chi}} = \frac{R/a^4}{Y/a^3} = \frac{R}{Y} \frac{1}{a} \propto a^{-1}, \quad (26)$$

with  $Y \equiv \rho_{\chi} a^3$  and  $R \equiv \rho_{\text{rad}} a^4$  both conserved in the absence of decay. The two densities therefore never cross; if the condensate did not decay, the universe would remain matter-dominated, with a radiation temperature today of only  $\sim 4 \times 10^{-10}$  eV. The pulse is thus a *seed* bath, not a reheating channel, and a genuine decay of the condensate (channel (iv)) is mandatory. (Earlier versions printed the same exponent  $\rho_{\text{rad}}/\rho_{\chi} \propto a^{-1}$  but described the ratio as *growing*, and consequently placed a crossing at  $a_{\text{reh}}/a_{\text{end}} \sim 10^{13}$  with  $T_{\text{reh}} \sim 4 \times 10^5$  GeV; that reading is incorrect, since the ratio decreases.)

(b) *Continuous conformal-anomaly decay* (channel (iv)): the rates and  $T_{\text{reh}}^{(\text{anom})} \simeq 2.1 \times 10^8$  GeV are as above.

***Combined picture.***

Reheating is driven *solely* by channel (iv); channel (i) provides only a seed radiation bath that dilutes away relative to the condensate (Eq. (26)) and does not by itself produce a radiation-dominated epoch. The two are nevertheless physically distinct—channel (i) is a one-time non-perturbative burst driven by the rapid metric change at the end of slow-roll, while channel (iv) is a continuous perturbative decay during the subsequent oscillatory phase—so no double-counting arises. Because the seed bath is negligible compared with the decay products,  $T_{\text{reh}}$  is set by channel (iv) alone to within  $\lesssim 10^{-3}$ . The fiducial value  $T_{\text{reh}} \simeq 10^9$  GeV (or  $2.1 \times 10^8$  GeV with the physical running coupling) is set by the continuous anomaly decay; under the locked  $N$  definition this corresponds to  $N \simeq 51$  (Table 1).

***Validity of the homogeneous-condensate approximation.***

The reheating estimates above treat the oscillating  $\chi$  field as a spatially homogeneous condensate. In a full preheating scenario, quantum fluctuations and non-linear resonance effects can fragment the condensate into inhomogeneous modes; when fragmentation is efficient, the uniform approximation breaks down and the energy transfer to radiation can be significantly faster than the perturbative estimate. For the present model, fragmentation is suppressed by two features: (i) the self-resonance parameter

$\lambda_{\text{self}} \sim 4 \times 10^{-11}$  (Sec. IV) is too small to drive efficient  $\chi$  fragmentation; (ii) the coupling to SM decay products is  $\leq 10^{-17}$ , so backreaction from produced radiation on the condensate is negligible. The homogeneous approximation is therefore well justified for the radiation-producing channel (iv). For the dark-matter fermion  $\psi$ , whose coupling  $g \simeq 1.0 \times 10^{-7}$  is  $\sim 10^{10}$  times larger than  $g_{\chi ff}$ , the homogeneous treatment is not sufficient and a lattice simulation is required (Discussion #3). Non-perturbative resonance effects could raise  $T_{\text{reh}}$  by one to two orders of magnitude, but a precise calculation requires a full lattice simulation. The perturbative fermion channel (channel (ii),  $\Gamma \sim 10^{-6}$  GeV) is subdominant; the graviton-pair channel (iii) is structurally closed; the open anomalous gauge channel is a robust feature of any scalar-tensor theory with  $F(\Phi)R/2$  coupling and SM embedding, analogous to the scalaron decay in  $R^2$  inflation [4].

#### *Parametric resonance into SM fermions and gauge bosons.*

During the oscillatory phase, the time-varying conformal factor  $\Omega(t) = \Phi(t)/\Phi_0$  induces a periodic modulation of SM particle masses:  $m_f(t) = m_f^{(0)}/\Omega(t)$  for Higgs-Yukawa fermions and  $\tilde{m}_W(t) = m_W/\Omega(t)$  for massive gauge bosons (Sec. VIII). In principle, such modulated mass terms can drive non-perturbative parametric resonance (Floquet/Mathieu instability), potentially competing with the anomalous decay channel (iv). However, the oscillation amplitude at the bottom of the potential is  $\delta\Phi/\Phi_0 \sim 10^{-111}$  (Sec. VII), so the fractional modulation  $\Delta m_f/m_f \sim 10^{-111}$  is utterly negligible after reheating. During the large-amplitude oscillation phase ( $\chi \sim M_{\text{Pl}}$ , immediately after the end of slow-roll), the modulation is  $\mathcal{O}(1)$ , but parametric resonance is driven by the *mass modulation*  $\delta m_f/m_f \sim \delta\Phi/\Phi_0 \sim \mathcal{O}(1)$  rather than by the Yukawa coupling  $g_{\chi ff}$ ; the correct Mathieu parameter [21] is  $q \sim (\delta\Phi/\Phi_0)^2 (m_f/m_\chi)^2 \sim (m_f/m_\chi)^2$ . For the top quark ( $m_t = 173$  GeV,  $m_\chi \simeq 3.25 \times 10^{13}$  GeV) this gives  $q \sim 2.8 \times 10^{-23}$ , and for lighter species it is exponentially smaller—over twenty orders below the broad-resonance threshold  $q \geq 0.1$ . The gauge-boson channel is likewise suppressed by  $\sim (m_W/M_{\text{Pl}})^2 \leq 10^{-33}$ . SM fermion and gauge-boson parametric resonance is therefore completely negligible for reheating; the dominant radiation sources remain the gravitational pulse (i) and the conformal-anomaly decay (iv). A full Floquet/lattice simulation of the dark-matter fermion  $\psi$  (whose coupling  $g \simeq 1.0 \times 10^{-7}$  is  $\sim 10^{10}$  times larger than  $g_{\chi ff}$ ) remains necessary for the absolute normalization of the relic abundance (Sec. V, Discussion #3).

#### **4.4 Post-reheating cosmology**

After reheating, the universe cools and follows standard  $\Lambda$ CDM. The thermalization rate  $\Gamma_{\text{therm}}/H \sim 10^7$  at  $T \sim 10^9$  GeV (see Discussion #17) ensures that the radiation bath reaches kinetic and chemical equilibrium within a few Hubble times. Non-thermal relics from gravitational particle production—including any high-energy tails of the fermion and gauge-boson spectra—are diluted by the dominant thermal bath; their residual non-thermal contributions to BBN light-element abundances are suppressed by the energy-density ratio  $\rho_{\text{non-th}}/\rho_{\text{rad}} \leq 10^{-13}$  and are far below current observational precision. The universe enters BBN with the standard radiation-dominated initial conditions, and the subsequent thermal history is indistinguishable from  $\Lambda$ CDM.

## 5 Fermionic dark matter from gravitational production

### 5.1 Setup: the Yukawa fermion $\psi$

The Yukawa fermion  $\psi$  already present in the action ( $\mathcal{L}_Y = -g\Phi\bar{\psi}\psi$ , Sec. II) is promoted to the dark matter. We take  $\psi$  to be a *Dirac, gauge-singlet* fermion with no Higgs portal ( $\bar{L}\tilde{H}\psi$  set to zero); both conditions are technically natural [11], since setting them to zero increases the symmetry. Its Einstein-frame mass is exact and  $\chi$ -independent:

$$m_\psi = g\Phi_0 = \frac{gM_{\text{Pl}}}{\sqrt{\xi}}, \quad (27)$$

a consequence of the conformal identity  $\Omega = \Phi/\Phi_0$  (Eq. (10)): the Yukawa term transforms as  $-g\Phi\bar{\psi}\psi\sqrt{-g} \rightarrow -g\Phi_0\bar{\tilde{\psi}}\tilde{\psi}\sqrt{-\tilde{g}}$  with  $\Omega^{1+3-4} = 1$  exactly, so the tree-level  $\chi\bar{\tilde{\psi}}\tilde{\psi}$  vertex vanishes identically for all  $\Phi$  and all  $g$  (Sec. II E).

### 5.2 Stability of $\psi$

The fermion  $\psi$  is stable under a discrete  $\mathbb{Z}_2^\psi$  symmetry under which  $\psi \rightarrow -\psi$  (all SM fields and  $\Phi$  are even). This discrete symmetry may be viewed as the low-energy remnant of a UV  $U(1)_D$  gauge symmetry spontaneously broken to  $\mathbb{Z}_2^\psi$ , and *if* realized as a gauge symmetry it is respected by all quantum-gravitational effects [24, 25]—in contrast to a global  $U(1)$ , which quantum gravity is expected to break. The  $\mathbb{Z}_2^\psi$  protection operates independently of the conformal decoupling:

- **Gauge-protected stability (conditional).** Under  $\mathbb{Z}_2^\psi$ , any operator connecting a single  $\psi$  to purely even SM final states ( $\psi \rightarrow \text{SM}$ ) is odd and strictly forbidden at all orders. Operators with an even number of  $\psi$  fields (e.g.  $\psi\bar{\psi}$  pair annihilation) are allowed but irrelevant at late times due to the low  $\psi$  abundance. The Dirac mass term  $m_\psi\bar{\psi}\psi$  is even and permitted, while single- $\psi$  decay is absolutely forbidden. No additional gauge fields or scalar degrees of freedom are required in the low-energy EFT; the discrete gauge nature of the symmetry is declared at the EFT level, with the UV  $U(1)_D \rightarrow \mathbb{Z}_2^\psi$  completion left implicit. For a discrete gauge symmetry to remain exact at the quantum level, the mixed  $\mathbb{Z}_2^\psi$ -gravitational and  $\mathbb{Z}_2^\psi$ -gauge anomalies must cancel [26, 27]; we assume this condition is satisfied by the UV completion (e.g. through the addition of anomaly-canceling heavy fermions at the  $U(1)_D$  breaking scale), which is a standard requirement for any discrete gauge symmetry [24] and does not affect the low-energy cosmology discussed here.
- **No renormalizable decay channel.** Being a gauge singlet,  $\psi$  has no renormalizable coupling to SM fields; with the  $\chi\psi\bar{\psi}$  vertex identically zero, the only residual interaction is gravitational (which respects  $\mathbb{Z}_2^\psi$ ). A Majorana mass term  $\frac{1}{2}m_M\psi^c\psi$  would violate  $\mathbb{Z}_2^\psi$  and is therefore excluded.

These two protection mechanisms operate at distinct levels and are not interchangeable:  $\mathbb{Z}_2^\psi$  forbids *any* single- $\psi$  decay operator—including those induced by quantum gravity (dimension-5  $\psi\sigma^{\mu\nu}F_{\mu\nu}$ , four-fermion operators, etc.)—at all orders,

provided the mixed anomalies cancel. The vanishing  $\chi\psi\bar{\psi}$  vertex, by contrast, is a tree-level kinematic feature of the conformal identity; it closes the perturbative reheating channel  $\chi \rightarrow \psi\bar{\psi}$  but would not *prevent*  $\psi$  decay through, e.g., a graviton-mediated  $\psi \rightarrow \nu\gamma$  loop if  $\mathbb{Z}_2^\psi$  were absent. Conversely, even if the  $\chi\psi\bar{\psi}$  vertex were non-zero (as in a generic scalar-tensor theory),  $\mathbb{Z}_2^\psi$  would still guarantee  $\psi$  stability, though perturbative reheating into  $\psi$  would then be open. The two mechanisms are therefore complementary:  $\mathbb{Z}_2^\psi$  provides the irreducible stability guarantee, while the conformally vanishing vertex ensures that reheating does not overproduce  $\psi$  through the perturbative channel, leaving gravitational production as the sole source (Sec. IV). This is the structural reason  $\psi$  can be dark matter where the condensate  $\chi$  cannot:  $\chi$  is a real scalar with no conserved number and a residual SM coupling  $g_{\chi ff} \sim m_f/M_{\text{Pl}}$  (Eq. (41)), so it decays in  $\tau_\chi \sim 10^{-27}$  s (graviton-mediated) and cannot survive to the present.

#### ***Indirect-detection prospects.***

Because  $\mathbb{Z}_2^\psi$  forbids  $\psi$  decay at all orders, there is no indirect-detection signal (gamma rays, neutrinos) from  $\psi$  annihilation or decay in the present universe. If quantum-gravity effects at the Planck scale were to induce  $\mathbb{Z}_2^\psi$ -breaking operators, the  $\psi$  decay lifetime would be  $\tau_\psi \sim M_{\text{Pl}}^2/m_\psi^3 \sim 10^{-20}$  s (for  $m_\psi \simeq 7.5 \times 10^{10}$  GeV), i.e. *instantaneous* on cosmological timescales. Therefore,  $\mathbb{Z}_2^\psi$  must be an exact, anomaly-free discrete gauge symmetry—any violation, even Planck-suppressed, would eliminate the dark-matter candidate entirely. This makes the UV completion (see discussion of anomaly-canceling heavy fermions in Sec. V) an essential, not optional, ingredient of the model. Under the assumption that such a UV completion exists and that  $\mathbb{Z}_2^\psi$  remains exact at all scales,  $\psi$  is stable with no indirect-detection signal.

#### ***Direct-detection prospects.***

The  $\psi$ -nucleon scattering cross section is dominated by graviton  $t$ -channel exchange, giving  $\sigma_{\psi N} \sim G_N^2 m_N^2 (\hbar c)^2 \sim 10^{-104}$  cm<sup>2</sup> (using  $G_N = 1/(8\pi M_{\text{Pl}}^2)$ ,  $m_N \sim 1$  GeV)—far below the XENONnT/LZ sensitivity floor ( $\sim 10^{-46}$  cm<sup>2</sup>). Higgs-exchange contributions (through radiatively generated  $\lambda_{\Phi H}$ ) are also negligible: the effective Yukawa  $y_\psi^{\text{eff}} \sim g_{\text{loop}} m_\psi/M_{\text{Pl}} \leq 10^{-29}$  yields  $\sigma_{\text{SI}} \leq 10^{-80}$  cm<sup>2</sup>. Direct detection is therefore irrelevant for this model. Direct-detection and accelerator searches are likewise insensitive to a fermion of mass  $\simeq 7.5 \times 10^{10}$  GeV coupled only gravitationally. The observational status of  $\psi$  is therefore that of a gravitationally produced superheavy relic whose absolute stability is conditional on the anomaly-free UV completion discussed above, testable only through its relic abundance (Sec. V) and its gravitational imprint on structure formation.

### **5.3 Closure of the perturbative reheating channel**

The same vanishing vertex closes the perturbative  $\chi \rightarrow \psi\bar{\psi}$  channel for *any*  $g$ , independent of kinematics. On the heavy branch  $m_\psi \gtrsim m_\chi/2$  the decay is *additionally*

kinematically forbidden, since

$$m_\chi \simeq 3.25 \times 10^{13} \text{ GeV} \approx 2H_{\text{inf}}, \quad 2m_\psi \geq 2H_{\text{inf}} \sim m_\chi, \quad (28)$$

so  $m_\chi \leq 2m_\psi$ : on that branch the closure is doubly protected. Vertex closure, by contrast, holds for *any*  $g$ . Kinematic closure additionally requires  $m_\psi \geq m_\chi/2$ , i.e.  $g \gtrsim \text{few} \times 10^{-5}$  at  $\xi = 11.1$ ; the lighter branch selected by the abundance below ( $g \simeq 1.0 \times 10^{-7}$ ,  $m_\psi \simeq 4.6 \times 10^{-3} H_{\text{inf}}$ , Sec. V) does not satisfy the kinematic condition and is closed by the vanishing vertex alone. The reheating channel of Sec. IV is therefore never reopened on either branch.

## 5.4 Production: gravitational particle production

The fermion  $\psi$  is produced at the end of inflation by a single physical mechanism—the non-adiabatic change of the spacetime metric at the end of slow-roll—which excites every minimally-coupled field from its vacuum (the same mechanism that produces the SM radiation of Sec. IV). Gravitational production of superheavy particles in the wimpzilla regime has been studied extensively [28–30], and the present model falls into this class. In the Einstein frame  $\psi$  is a free Dirac field of constant mass  $m_\psi = g\Phi_0$  with no  $\chi$  coupling (Sec. V), so its production is governed entirely by the time-dependent Friedmann background; for  $m_\psi \leq H_{\text{inf}} \simeq 1.64 \times 10^{13} \text{ GeV}$  the production is near its peak (wimpzilla regime [22, 23]), with number density

$$n_\psi(a_{\text{end}}) \sim H_{\text{inf}}^3 e^{-2\pi m_\psi/H_{\text{inf}}} \quad (29)$$

up to  $\mathcal{O}(1)$  factors, where  $n_\psi^{\text{tot}} \equiv n_\psi + n_{\bar{\psi}}$  denotes the total number density of particles plus antiparticles (gravitational production is symmetric under  $\psi \leftrightarrow \bar{\psi}$ ). The exponent follows from the exact Bogoliubov coefficient of a Dirac field in de Sitter,  $|\beta_k|^2 = 1/(e^{2\pi m_\psi/H_{\text{inf}}} + 1)$  with index  $\nu = \frac{1}{2} \mp i m_\psi/H_{\text{inf}}$  [31, 32]; the factor of 2 relative to a single Boltzmann factor arises because the exact coefficient of a single mode contains two suppressing factors:  $|\Gamma(\frac{1}{2} + i\mu)|^2 = \pi/\cosh(\pi\mu) \simeq 2\pi e^{-\pi\mu}$  and the Bunch–Davies normalization of the mode function,  $e^{-\pi m_\psi/(2H_{\text{inf}})}$ , which enters squared in  $|\beta_k|^2$ . This process does *not* use the vanishing perturbative  $\chi\psi\bar{\psi}$  vertex.

### *Scope of the estimate.*

Equation (29) is the de Sitter-era result and should be regarded as an upper bound on the suppression rather than as the final answer. Two effects go beyond it. First, for a fermion the massless limit is conformal, so gravitational production is suppressed by  $m_\psi$ : the production rate *rises* with  $m_\psi$  up to  $m_\psi \sim m_\chi$  rather than peaking at  $H_{\text{inf}}$ , and the reheating-era contribution from the oscillating scale factor  $a(t)$  is comparable to (and for  $m_\psi \gtrsim H_{\text{inf}}$  dominates over) the slow-roll contribution, without the exponential factor of Eq. (29) [31, 33]. Second, in the present model the condensate mass satisfies  $m_\chi/H_{\text{inf}} \simeq 1.99$ , i.e. the post-inflationary phase is not a fast-oscillation regime ( $2\pi H_{\text{inf}}/m_\chi \simeq 3.2$   $e$ -folds per oscillation), so the usual parametric-resonance treatment does not apply either. The abundance quoted below therefore carries an

$\mathcal{O}(1)$  (possibly larger) normalization uncertainty that only a Floquet/lattice computation can remove, exactly as stated in the Discussion. In what follows we go one step beyond Eq. (29): the abundance matching is performed with the numerically measured spectrum  $n_\psi(m_\psi)$  obtained by integrating the mode equation across the transition (Appendix D0), which for  $m_\psi \gtrsim H_{\text{inf}}$  is only weakly dependent on  $m_\psi$  rather than exponentially suppressed, and therefore admits a light solution as well.

### ***Jordan-frame cross-check.***

The same production admits an equivalent Jordan-frame description in which the metric is rescaled by the time-varying  $\Omega = \Phi/\Phi_0$  and the fermion mass  $m_\psi(t) = g|\Phi(t)|$  is time-dependent. The adiabaticity parameter at the post-inflationary oscillation amplitude  $\Phi_{\text{osc}} \sim 0.8 \Phi_0$  is

$$\frac{\dot{m}_\psi}{m_\psi^2} \sim \frac{m_\Phi \Phi_{\text{osc}}}{g \Phi_0^2} \sim \frac{4.4 \times 10^{-4}}{g}, \quad (30)$$

so adiabaticity fails for  $g \lesssim 10^{-4}$ , and at the abundance-matched value  $g \simeq 1.0 \times 10^{-7}$  the parameter (30) reaches  $\sim 4 \times 10^3$ , i.e. strongly non-adiabatic  $\psi\bar{\psi}$  pair production—consistent with the light branch found below, which sits at  $m_\psi \ll H_{\text{inf}}$ . This Jordan-frame picture is the *conformal image* of the Einstein-frame gravitational production above, not an independent channel: both are driven by the same single  $\Phi(t)$  dynamics through the conformal relation  $\tilde{g}_{\mu\nu} = \Omega^2(\Phi)g_{\mu\nu}$ , so they should not be added. The Jordan-frame adiabaticity check serves only as a consistency cross-check on the location of the production peak; the abundance estimate below uses the Einstein-frame result (29) alone. The precise  $m_\psi$  that yields the observed abundance is not computable from this parametric estimate alone; it is obtained below from the mode-equation integration, with the residual normalization requiring a Floquet/lattice treatment (Discussion).

## **5.5 Abundance: mechanism and window**

After production  $\psi$  is non-interacting and redshifts as matter,  $n_\psi \propto a^{-3}$ , decoupled from both the decaying condensate and the radiation bath. Diluting  $n_\psi(a_{\text{end}}) \sim H_{\text{inf}}^3$  by the entropy dilution factor  $(a_{\text{end}}/a_0)^3$  gives, parametrically,

$$\Omega_\psi \sim \frac{m_\psi H_{\text{inf}}^3 e^{-2\pi m_\psi/H_{\text{inf}}} (a_{\text{end}}/a_0)^3}{\rho_c}. \quad (31)$$

The dilution factor scales as  $(a_{\text{end}}/a_0)^3 \propto T_{\text{reh}}$  (from  $a_{\text{end}}/a_{\text{reh}} \propto T_{\text{reh}}^{4/3}$  during matter-like condensate domination and  $a_{\text{reh}}/a_0 \propto T_{\text{reh}}^{-1}$  during radiation domination), so for the fiducial  $T_{\text{reh}} \sim 10^9 \text{ GeV}$  we have  $(a_{\text{end}}/a_0)^3 \simeq 1.02 \times 10^{-92}$ . The exponential suppression lets  $\Omega_\psi$  be tuned through  $\mathcal{O}(1)$  near  $m_\psi \sim H_{\text{inf}}$ ; the precise matching point within

$$m_\psi \simeq 3.1 H_{\text{inf}} \simeq 5.1 \times 10^{13} \text{ GeV}, \quad g = \frac{m_\psi}{\Phi_0} \simeq 6.9 \times 10^{-5}, \quad (32)$$

gives  $\Omega_\psi = \Omega_{\text{DM}} \simeq 0.265$  at  $T_{\text{reh}} = 10^9$  GeV. The Yukawa coupling  $g$  is a free parameter of the model, fixed a posteriori by the dark-matter relic abundance; it is not predicted from first principles, in the same sense that  $\xi$  and  $\lambda_0$  are fitted to  $A_s$  and the  $r$  bound. Because  $(a_{\text{end}}/a_0)^3 \propto T_{\text{reh}}$ , a one-order-of-magnitude shift in  $T_{\text{reh}}$  moves the matching  $g$  by only 1.13 (for  $u \equiv 2\pi m_\psi/H_{\text{inf}} \simeq 19$ ,  $d \ln g = d \ln T/(u-1)$ )—a weak sensitivity specific to this exponential branch, to be contrasted with the light branch found below, where  $m_\psi n_\psi \propto m_\psi^2$  gives instead  $g \propto T_{\text{reh}}^{-1/2}$ , a factor  $\sqrt{10}$ ; across the BBN-allowed  $T_{\text{reh}} \in [10^{-2}, 2.6 \times 10^{15}]$  GeV, the matching  $g$  varies by a factor of a few, so the gravitational/anomaly reheating estimate  $T_{\text{reh}} \sim 10^9$  GeV is what selects  $g \simeq 7 \times 10^{-5}$ , not a narrow prediction. This matching assumes that the de Sitter exponential (29) dominates the production. As noted above, the reheating-era contribution is *not* exponentially suppressed, so  $g \simeq 6.9 \times 10^{-5}$  should be read as the matching point *within that framework* rather than as a prediction; the full mode-equation calculation is deferred (Discussion). Note also that  $\Omega_\psi$  as a function of  $g$  has a maximum at  $g = (2\pi)^{-1} H_{\text{inf}}/\Phi_0 \simeq 3.6 \times 10^{-6}$ ; the matching point is on the large- $g$  branch, and the lower part of the interval  $g \sim 10^{-5}$ – $10^{-4}$  overproduces  $\psi$  within this framework; the mode-equation result of the next paragraph supersedes this picture.

***One light branch, not two competing branches.***

Equation (31) is non-monotonic in  $g$ , so the condition  $\Omega_\psi = \Omega_{\text{DM}}$  does not by itself single out Eq. (32). Because a massless Dirac field is conformally invariant, gravitational production switches off as  $m_\psi \rightarrow 0$ ; hence  $\Omega_\psi(g) \rightarrow 0$  on the light side and a second solution must exist below the maximum at  $g = (2\pi)^{-1} H_{\text{inf}}/\Phi_0 \simeq 3.6 \times 10^{-6}$ . Replacing the de Sitter exponential (29) by the actual mode-equation spectrum across the end-of-inflation transition (Appendix D0), and fixing the model’s own anomaly-channel temperature  $T_{\text{reh}} = 2.1 \times 10^8$  GeV, we find that  $\Omega_\psi = \Omega_{\text{DM}}$  is met at  $g \simeq 1.0 \times 10^{-7}$ , i.e.  $m_\psi \simeq 4.6 \times 10^{-3} H_{\text{inf}} \simeq 7.5 \times 10^{10}$  GeV, where  $n_\psi/H_{\text{inf}}^3 \simeq 1.4 \times 10^{-5}$  and  $m_\psi n_\psi \propto m_\psi^2$ . Over the same spectrum the heavy side is *not* a competing solution:  $\Omega_\psi$  rises monotonically with  $m_\psi$  for  $m_\psi \gtrsim H_{\text{inf}}$  (we measure  $\Omega_\psi \approx 1.6 \times 10^3$ – $3.0 \times 10^3$  over  $m_\psi/H_{\text{inf}} = 3$ – $12$ ), so Eq. (32) overproduces  $\psi$  by  $\sim 6 \times 10^3$  and every heavier value overproduces at least as much. The heavy side becomes viable only if  $T_{\text{reh}}$  is lowered by some four orders of magnitude—we find  $T_{\text{reh}} \simeq 2$ – $3 \times 10^4$  GeV over  $m_\psi/H_{\text{inf}} \in [1, 12]$ —which the conformal-anomaly channel of Sec. IV does not provide. Within the transition calculation the abundance matching is therefore unique, and we read  $g \simeq 1.0 \times 10^{-7}$  as an upper bound on the required coupling, since the oscillating-condensate branch (not contained in that calculation) can only add to the production. Equation (32) is retained as the reference point of the exponential framework rather than as a prediction; the absolute normalization still requires the lattice calculation discussed below.

The mechanism is viable and, within the transition calculation, matches the observed abundance on the light branch at  $g \simeq 1.0 \times 10^{-7}$ ; the absolute normalization and the split of the condensate energy between radiation and the stable  $\psi$  component require a full Floquet/lattice simulation of preheating (Discussion).

## 5.6 Parallel option: topological defects

Gravitational freeze-in of a Yukawa fermion is not the only way this scalar sector can source dark matter. Non-perturbative  $\mathbb{Z}_2$ -breaking dynamics during preheating can leave **topological defects** (domain-wall remnants, textures, or related configurations) with approximately conserved defect energy  $E_{\text{def}}$  and comoving number, so that

$$n_{\text{def}} \propto a^{-3} \implies \rho_{\text{DM}}^{\text{def}} = E_{\text{def}} n_{\text{def}} \propto a^{-3}, \quad (33)$$

which is the required dust-like scaling, familiar from defect dark-matter scenarios [43]. This channel is a legitimate phenomenological alternative consistent with the same field content. It does *not* supply a first-principles  $\Omega_{\text{DM}}$  without lattice input (formation efficiency, bias, and survival). We therefore list defect dark matter as a parallel channel, not as a completed calculation.

In contrast, a *homogeneous* condensate cannot play this role. From  $\ddot{\Phi} + 3H\dot{\Phi} + V' \simeq 0$  near a quadratic minimum one has  $\dot{\Phi} \propto a^{-3}$  and hence kinetic energy  $\propto a^{-6}$ , far too fast to remain a 27% component today; the condensate must decay to reheat the universe (Sec. IV). The same  $a^{-6}$  argument underlies the exclusion of single-field dynamical dark energy in Sec. VI.

## 5.7 Compatibility with reheating

The energy density stored in  $\psi$  at production is  $\rho_{\psi}(a_{\text{end}}) = m_{\psi} n_{\psi} \simeq 9.5 \times 10^{44} \text{ GeV}^4$  at the fiducial  $T_{\text{reh}} = 10^9 \text{ GeV}$  (it scales as  $T_{\text{reh}}^{-1}$  at fixed  $\Omega_{\psi}$ ), i.e.  $\rho_{\psi}/\rho_{\text{end}} \simeq 5.8 \times 10^{-19}$ , so  $\psi$  does not contest the condensate's dominance; reheating proceeds through the conformal-anomaly decay channel (iv) of Sec. IV, which provides  $T_{\text{reh}} \lesssim 10^9 \text{ GeV}$ ; the initial gravitational pulse supplies only a seed bath and cannot reheat the universe on its own (Sec. IV). Requiring  $\Omega_{\psi} = 0.265$  fixes  $\rho_{\psi}/\rho_{\text{rad}} \simeq 5.8 \times 10^{-19}$  at reheating (the same number, since  $\rho_{\text{rad}}(T_{\text{reh}}) = \rho_{\text{cond}}(a_{\text{reh}})$ ), so  $\psi$  is a spectator to the thermal history and BBN is unaffected. The condensate and  $\psi$  thus have disjoint fates—the condensate decays to reheat,  $\psi$  survives as dark matter—resolving the tension that would arise if the condensate itself had to be both the reheating fuel and the dark matter.

# 6 Late-time dark energy: frozen $\Lambda$ CDM

## 6.1 Late-time behavior

Expanding  $V_E(\varphi)$  near  $\varphi \rightarrow 0$ :  $V_E(\varphi) \simeq (\lambda_0 M_{\text{Pl}}^4/\xi^2)\varphi^2 + V_c(1 - 4\varphi + 8\varphi^2 + \dots)$ . The canonical-field mass is Eq. (12),  $m_{\chi} \sim 3.25 \times 10^{13} \text{ GeV}$  (locked  $N = 50$ ). With  $m_{\chi} \gg H_0 \sim 10^{-42} \text{ GeV}$ , the field is underdamped and frozen at the minimum.

The post-inflationary condensate decays during preheating through conformal-anomaly channel (Sec. IV), completing reheating; it does *not* survive as dark matter. The cold dark matter is instead the Dirac fermion  $\psi$  of Sec. V, gravitationally produced and redshifting as  $a^{-3}$ , a separate  $w = 0$  fluid decoupled from both the condensate and the radiation bath.

Once reheating ends, the field settles at  $\langle\Phi\rangle = \Phi_0$  and  $V_J(\Phi_0) = V_c$  persists as a cosmological constant. The  $V_c$  contribution alone gives a cosmological-constant equation of state,

$$w_0 = -1, \quad w(z) = -1 \quad (\forall z), \quad (34)$$

up to residual-condensate corrections bounded in App. D5 (the dominant correction  $\Delta w \leq 10^{-10}$  for  $T_{\text{reh}} \geq 1 \text{ GeV}$ , consistent with our gravitational-reheating estimate). The  $w = -1$  result follows because  $V_c$  is the constant part of the potential: it contributes  $T_{\mu\nu}^{(\text{CC})} = -V_c g_{\mu\nu}$  and drops out of the scalar equation of motion  $\ddot{\chi} + 3H\dot{\chi} + m_\chi^2\chi = 0$ , so it evolves independently of both the decaying condensate and the stable  $\psi$  fluid. The  $\psi$  dark matter is a separate  $w = 0$  cold-dark-matter fluid (Sec. V); together with the  $V_c$  cosmological constant the late-time model is effectively  $\Lambda$ CDM to extraordinary precision (residual corrections bounded in App. D5; for  $T_{\text{reh}} \geq 1 \text{ GeV}$  these are  $\leq 10^{-10}$ ), with  $w_{\text{DE}} = -1 + \Delta w$ ,  $|\Delta w| \ll 1$ , and no evolution or phantom crossing possible.

### Structure growth.

The tree-level conformal decoupling ( $g_{\chi ff} = 0$  for  $\psi$ , Sec. II E) implies that scalar-field perturbations  $\delta\chi$  do not source matter density perturbations in the  $\psi$  sector; for SM matter the residual couplings are tiny. The growth equation  $\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G\rho_m\delta_m = 0$  is identical to  $\Lambda$ CDM at leading order. The effective coupling induced by graviton loops ( $\sim 10^{-11}g$ , see loop discussion in Sec. II E) is far too weak to produce a measurable deviation in  $f\sigma_8$  ( $\Delta f/f \sim 10^{-22}$ ). The model therefore predicts  $f\sigma_8(z)$  indistinguishable from  $\Lambda$ CDM and automatically compatible with DESI, eBOSS, and future Stage-IV surveys.

## 6.2 Residual motion bound

Three sources of residual  $\chi$  motion that could perturb the frozen minimum are bounded (full derivations in App. D5):

- **Ricci driving:**  $\Delta w \sim (H_0/m_\chi)^2 \sim 2 \times 10^{-111}$ .
- **Quantum fluctuations:**  $\Delta w \sim H_0^4/V_c \sim 2 \times 10^{-121}$ .
- **Residual condensate tail:** the oscillating condensate redshifts as  $a^{-3}$  (pressureless dust,  $w = 0$ ), not as dark energy. Its relic density is subject to the overclosure constraint  $\rho_{\text{cond}}(a_0) \leq \rho_{\text{crit}}$ , i.e.  $\rho_{\text{cond}}(a_0)/V_c \leq 1/\Omega_\Lambda$  (or  $\leq \Omega_{\text{DM}}/\Omega_\Lambda \simeq 0.39$  if the condensate is required not to exceed the observed dark-matter budget). This bound is *satisfied*, but only because the condensate decays: an absolutely stable condensate would give  $\rho_{\text{cond}}(a_0)/V_c \simeq 7 \times 10^{17}$ , i.e. it would overclose the universe by  $\sim 10^{18}$ . What makes the bound hold is the decay-completeness condition  $\Gamma t_0 \gg 1$ , satisfied by an enormous margin for any  $T_{\text{reh}} \gtrsim 10^{-2} \text{ GeV}$  (e.g.  $\Gamma t_0 \simeq 3 \times 10^{20}$  at the BBN floor and  $\simeq 3 \times 10^{42}$  at  $T_{\text{reh}} = 10^9 \text{ GeV}$ ), so that the residual at  $a_0$  is strictly zero. A dedicated integration of the two-fluid Boltzmann equations (App. D5c) confirms this: tracking the condensate as  $Y \equiv \rho_{\text{cond}}a^3$  and the radiation as  $R \equiv \rho_{\text{rad}}a^4$ , the residual today vanishes for every  $\Gamma \neq 0$  across  $T_{\text{reh}} \in [10^{-2}, 10^{15}] \text{ GeV}$ . The condensate therefore contributes to  $\Omega_m$  only through its decay products, not to  $\Delta w_{\text{DE}}$ .

The first two are far below the DESI/Euclid  $3\sigma$  sensitivity of  $\Delta w \sim 10^{-3}$ ; the third is bounded by the overclosure constraint and the rapid damping  $\tau \sim m_\chi^{-1} \sim 10^{-38} \text{ s}$  (App. D4), which jointly render the condensate cosmologically irrelevant at late times.

The condensate decays completely during preheating, so it leaves no residual component that could perturb the frozen minimum. The dark matter  $\psi$  is a separate  $w = 0$  cold-dark-matter fluid and does not enter the dark-energy equation of state:  $w_{\text{DE}} = -1$  up to residual-condensate corrections bounded in App. D5 (negligible for  $T_{\text{reh}} \geq 1 \text{ GeV}$ ). The corresponding total effective equation of state  $w_{\text{tot}} = -V_c/(\rho_\psi + V_c)$  is *not* the quantity DESI/BAO/SNe constrain—standard  $\Lambda\text{CDM}$  has  $w_{\text{tot}} \approx -0.69$  and is likewise unexcluded.

### 6.3 Calibration and fine-tuning

$V_c$  is calibrated by  $\Omega_\Lambda = 0.683$ :

$$V_c = \Omega_\Lambda \cdot 3H_0^2 M_{\text{Pl}}^2 \approx 2.7 \times 10^{-47} \text{ GeV}^4. \quad (35)$$

The  $10^{110}$  hierarchy between  $V_0 \sim 10^{63} \text{ GeV}^4$  and  $V_c$  is spanned by the conformal factor  $e^{-4\varphi}$  but *not* dynamically explained; this is the cosmological constant problem, shared with all  $\Lambda\text{CDM}$  models. This model reproduces a late-time cosmological constant with  $w = -1$  (up to corrections  $\leq 10^{-10}$  for  $T_{\text{reh}} \geq 1 \text{ GeV}$ ; see App. D5), but does not address the core fine-tuning puzzles. Two distinct tuning problems remain: (i) why  $\Phi_0 = M_{\text{Pl}}/\sqrt{\xi} \sim 10^{17} \text{ GeV}$  sets the Planck/electroweak hierarchy—the VEV is a fixed point of the potential by construction but its magnitude is not dynamically explained; (ii) why  $V_c \sim 10^{-47} \text{ GeV}^4$  is  $10^{110}$  times smaller than the inflationary scale  $V_0 \sim 10^{63} \text{ GeV}^4$ —the conformal factor  $e^{-4\varphi}$  maps  $V_c$  from the Jordan to the Einstein frame but the tiny value of  $V_c$  itself is not derived from a symmetry or dynamical principle [34]. Moreover, the heavy fields  $\chi$  ( $m_\chi \sim 10^{13} \text{ GeV}$ ) and  $\psi$  ( $m_\psi \simeq 7.5 \times 10^{10} \text{ GeV}$ ) contribute radiative corrections  $\delta V_c \sim m^4/(16\pi^2) \sim 10^{41}\text{--}10^{52} \text{ GeV}^4$ , exacerbating the fine-tuning by an additional  $\sim 10^{88}\text{--}10^{99}$  orders relative to the observed  $V_c$ ; this makes the cosmological constant problem quantitatively more severe in this model than in the Standard Model alone, an issue shared with any theory containing heavy scalar or fermion degrees of freedom and deferred to a UV completion.  $V_c$  is an empirical observational input, not dynamically derived. During reheating, the minimum at  $\Phi = \Phi_0$  is protected against thermal and curvature perturbations: the field mass  $m_\Phi \sim 2.7 \times 10^{14} \text{ GeV}$  exceeds the reheating temperature  $T_{\text{reh}} \sim 10^9 \text{ GeV}$  by five orders of magnitude, thermal mass corrections  $\delta m_\Phi^2 \sim g^2 T^2$  are  $\leq 10^{-20} m_\Phi^2$ , and curvature-driven displacements  $\sim \xi \Phi_0 R/m_\Phi^2$  are  $\leq 10^{-13}$ ; the minimum remains stable throughout all post-inflationary epochs. Future DESI [35]/Euclid compressing  $\sigma(w_0)$  to  $\sim 10^{-3}$  could test  $w_0 = -1$  at  $> 3\sigma$ ; a detection of  $w_0 \neq -1$  would exclude this model and  $\Lambda\text{CDM}$  alike.

## 7 Evolving active parameter

Although the scalar field is frozen at  $\Phi = \Phi_0$  to extraordinary precision (Sec. VI), the residual Ricci-driven displacement and the corresponding drift of fundamental

constants provide a consistency check that the induced-gravity framework does not generate observably large violations of the equivalence principle or time-variation of couplings. All effects derived in this section lie  $> 90$  orders below current experimental sensitivity; the section may be skipped on a first reading without loss of continuity.

In FRW, the linearized field equation reads

$$\delta\ddot{\Phi} + 3H\delta\dot{\Phi} + (2\lambda_0\Phi_0^2 - \xi R)\delta\Phi = \xi\Phi_0 R. \quad (36)$$

In the quasi-static limit:

$$\frac{\Delta\mathcal{A}}{\mathcal{A}} = + \frac{\xi R}{\lambda_0\Phi_0^2}. \quad (37)$$

With  $R_0 \approx 9.2H_0^2$ :  $\Delta\mathcal{A}/\mathcal{A}|_0 \approx +6.3 \times 10^{-111}$ , far below foreseeable sensitivity.

***Fine-structure-constant drift.***

$\Delta\alpha/\alpha = (de/d\Phi)(\Phi_0/e) \cdot \Delta\mathcal{A}/\mathcal{A}$ . Three scenarios: (i)  $e$  independent of  $\Phi \Rightarrow \Delta\alpha/\alpha = 0$ ; (ii) mass-coupling-dominance  $\Rightarrow |\Delta\alpha/\alpha| \sim 10^{-110}$ ; (iii) logarithmic dependence  $\Rightarrow$  even smaller. Scenario (ii) lies  $\sim 90$  orders below current sensitivity.

## 8 Standard Model embedding and renormalization-group stability

The analysis of Sec. II E demonstrates that the direct Yukawa coupling  $-g\Phi\bar{\psi}\psi$  is exactly decoupled by the conformal identity  $\Omega = \Phi/\Phi_0$ . This section constructs the minimal SM extension— $\Phi$  couples to SM fields only through the metric (no direct Higgs portal)—and verifies RG stability.

### 8.1 Jordan-frame action with Standard Model fields

The total Jordan-frame action is the sum of the scalar-tensor sector (1) and the SM Lagrangian in curved spacetime:

$$S_{\text{SM}} = \int d^4x \sqrt{-g} \left[ -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \bar{\psi}_f i\gamma^\mu D_\mu \psi_f + |D_\mu H|^2 - V_{\text{SM}}(H) - y_f \bar{\psi}_L \psi_R H + \text{h.c.} \right], \quad (38)$$

where  $F_{\mu\nu}^a$  are SM gauge field strengths,  $\psi_f$  denotes SM fermion families,  $H$  is the Higgs doublet with  $V_{\text{SM}}(H) = -\mu_H^2 |H|^2 + \lambda_H |H|^4$ , and  $y_f$  are the Yukawa couplings.  $\Phi$  is a gauge singlet with no renormalizable direct coupling to SM fields; setting the Higgs portal  $\lambda_{\Phi H} \Phi^2 |H|^2$  to zero is technically natural [11] (the limit  $\lambda_{\Phi H} \rightarrow 0$  enhances the symmetry by decoupling  $\Phi$  from the Higgs sector). Radiative corrections induce  $\delta\lambda_{\Phi H}$  only through graviton exchange, giving  $\delta\lambda_{\Phi H} \sim \xi\lambda_0/(16\pi^2) \times (m_\Phi^2/M_{\text{Pl}}^2) \sim 10^{-15}$ , far below any observable threshold. All SM matter couples to  $\Phi$  only through the Jordan-frame metric.

## 8.2 Gauge sector

Under  $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$ , the gauge kinetic term transforms as

$$-\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} \sqrt{-g} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} \sqrt{-\tilde{g}}, \quad (39)$$

because  $g^{\mu\rho}g^{\nu\sigma}F_{\mu\nu}F_{\rho\sigma}$  is conformally invariant in four dimensions. Gauge fields are not rescaled; SM gauge couplings  $g_1, g_2, g_3$  are unchanged and there is no tree-level  $\chi$ -gauge vertex. At the 1-loop level, however, the conformal (trace) anomaly generates an effective  $\chi$ -gauge coupling  $\mathcal{L}_{\text{eff}} \supset (b_i \alpha_i / 4\pi)(\chi / (M_{\text{Pl}} \sqrt{6 + 1/\xi})) F_i^2$ , opening the  $\chi \rightarrow gg, \gamma\gamma$  decay channels at  $\Gamma_{\text{anom}} \sim \mathcal{O}(10^{-1})$  GeV. Because this is a continuous perturbative decay it is the channel that actually reheats the universe, giving the fiducial  $T_{\text{reh}} \lesssim 10^9$  GeV (Sec. IV); the gravitational pulse (i) is a single initial burst that dilutes faster than the condensate and cannot do so on its own. The tree-level decoupling of  $\chi$  from gauge bosons remains exact for the purpose of fifth-force bounds, which are set by the scalar's large mass.

### *Massive gauge bosons.*

After electroweak symmetry breaking, the  $W^\pm$  and  $Z$  bosons acquire mass. Under the conformal transformation  $g_{\mu\nu} = \Omega^{-2} \tilde{g}_{\mu\nu}$ , the mass term transforms as  $\frac{1}{2} m_W^2 g_{\mu\nu} W^\mu W^\nu \sqrt{-g} \rightarrow \frac{1}{2} (m_W / \Omega)^2 \tilde{g}_{\mu\nu} \tilde{W}^\mu \tilde{W}^\nu \sqrt{-\tilde{g}}$  (with  $\tilde{W}_\mu \equiv \Omega^{-1} W_\mu$ ), so the effective Einstein-frame mass is  $\tilde{m}_W = m_W / \Omega$ . Since  $\Omega = \Phi / \Phi_0 \rightarrow 1$  after reheating (the field settles at  $\Phi = \Phi_0$  with residual displacement  $\delta\Phi / \Phi_0 \sim 10^{-111}$ ; Sec. VII), the standard  $W$  and  $Z$  masses are recovered to extraordinary precision in the late-time SM. During inflation,  $m_W$  is exponentially suppressed (as is the entire electroweak scale), but this is irrelevant since the Higgs field is displaced from its minimum and electroweak symmetry is not yet broken.

## 8.3 Fermion sector

The curved-space fermion kinetic term is conformally invariant after  $\psi = \Omega^{3/2} \tilde{\psi}$  [23]. Higgs Yukawa terms transform as

$$y_f \bar{\psi}_L \psi_R H \sqrt{-g} \longrightarrow y_f \Omega^{3/2} \tilde{\psi}_L \Omega^{3/2} \tilde{\psi}_R (\tilde{H} / \Omega) \Omega^{-4} \sqrt{-\tilde{g}} = y_f \Omega^{-2} \tilde{\psi}_L \tilde{\psi}_R \tilde{H} \sqrt{-\tilde{g}}, \quad (40)$$

where  $\tilde{H} \equiv \Omega H$ . Unlike the direct  $-g\Phi\bar{\psi}\psi$  coupling, where  $\Phi \propto \Omega$  cancels all factors, the SM Higgs Yukawa leaves a residual factor  $\Omega^{-2}$ . The Yukawa fermion  $\psi$  of Sec. V is taken to be a Dirac, gauge-singlet fermion with no Higgs portal ( $\bar{L}\tilde{H}\psi$  set to zero, technically natural [11]); it is therefore distinct from the SM fermions  $\psi_f$  treated here, has no residual  $\chi$  coupling, and is stable under the discrete  $\mathbb{Z}_2^\psi$  symmetry (Sec. V). The physical Einstein-frame fermion mass is  $m_f = m_f^{(0)} / \Omega = m_f^{(0)} e^{-\varphi}$ : the  $\Omega^{-2}$  of Eq. (40) multiplies the field  $\tilde{H}$ , whose expectation value  $\langle \tilde{H} \rangle = \Omega v$  carries a compensating  $\Omega$ .

Hence the effective  $\chi$ -fermion coupling is

$$g_{\chi ff} \simeq \frac{m_f^{(0)}}{M_{\text{Pl}} \sqrt{6 + 1/\xi}} \sim 10^{-22} - 10^{-17}, \quad (41)$$

many orders below the Cassini bound [12]. The range  $\sim 10^{-22} - 10^{-17}$  reflects the fermion-mass hierarchy: the lower end corresponds to the electron ( $m_e \sim 0.5$  MeV) and the upper end to the top quark ( $m_t \sim 173$  GeV); even at the top end the coupling is far below the solar-system bound, and the corresponding decay is negligible for reheating (see Sec. IV, channel (ii)). The corresponding decay rate  $\Gamma(\chi \rightarrow f\bar{f})/m_\chi \leq 5 \times 10^{-35}$  (quoted for the top, the largest coupling) is negligible for reheating. For comparison,  $\Gamma_{\chi \rightarrow t\bar{t}} \sim 1.5 \times 10^{-21}$  GeV versus the anomaly width  $\Gamma_{\text{anom}} \sim \mathcal{O}(10^{-1})$  GeV of channel (iv), a  $\sim 7 \times 10^{19} \times$  suppression; the direct  $\chi ff$  coupling is therefore completely irrelevant for reheating.

## 8.4 Higgs sector and field-space metric

After rescaling  $H = \tilde{H}/\Omega$ , the Einstein-frame kinetic terms for  $\chi$  and  $\tilde{H}$  read

$$\mathcal{L}_{\text{kin}} = \frac{1}{2}(\partial\chi)^2 + \frac{1}{2}e^{-2\varphi}|D_\mu\tilde{H}|^2, \quad \frac{d\varphi}{d\chi} = \frac{1}{M_{\text{Pl}}\sqrt{6 + 1/\xi}}. \quad (42)$$

At late times  $\varphi \rightarrow 0$ , the Higgs kinetic term reduces to the canonical SM form. Since  $\varphi$  is frozen with  $m_\chi \gg H_0$ , all  $\chi$ - $\tilde{H}$  mixing is kinematically inaccessible; the SM is recovered exactly.

During inflation ( $\varphi \gg 1$ ), the Higgs mass term is exponentially suppressed by  $e^{-4\varphi}$ , so  $H$  is effectively massless and does not affect inflationary dynamics. Isocurvature perturbations from Higgs- $\Phi$  field-space curvature are estimated to be well below Planck sensitivity for  $\varphi \geq 5$  in generic Higgs-inflation setups [36]; a dedicated computation of the isocurvature power spectrum for the present model's specific parameters ( $\xi = 11.1$ ,  $\lambda_0 \simeq 6.70 \times 10^{-8}$  at locked  $N = 50$ ,  $\xi_h = 0$ ) and a direct comparison with Planck 2018 isocurvature bounds [16] have not been performed and are deferred to a future multifield analysis. A Higgs non-minimal coupling  $\xi_h|H|^2R$  is in principle allowed in the Jordan frame; if present, it would induce an Einstein-frame term  $\xi_h|\tilde{H}|^2\tilde{R}/\Omega^2$ , contributing to the field-space metric during inflation. For simplicity we set  $\xi_h = 0$ ; a nonzero  $\xi_h \sim \mathcal{O}(1)$  would shift the inflationary observables by  $\leq 10^{-4}$  [3] and does not affect our conclusions.

### *Electroweak phase transition.*

Since  $T_{\text{reh}} \sim 10^9$  GeV  $\gg v$ , the electroweak symmetry is restored after reheating and broken when the Universe cools below  $T \sim 100$  GeV. With  $\xi_h = \lambda_{\Phi H} = 0$ , the  $\Phi$ -Higgs mixing vanishes at tree level and the electroweak phase transition proceeds as in the SM (a smooth crossover [37]), unmodified by the inflaton background. If loop effects were to induce a small  $\lambda_{\Phi H}$ , the transition dynamics could in principle be altered; a quantitative study of electroweak baryogenesis in this framework is deferred to future work.

## 8.5 Renormalization-group stability of $\xi$ and $\lambda_0$

For a real scalar singlet with a Yukawa fermion  $\psi$ , the 1-loop  $\beta$ -functions in the  $\overline{\text{MS}}$  scheme are [22, 38]

$$\beta_{\lambda_0} \equiv \frac{d\lambda_0}{d\ln\mu} = \frac{9\lambda_0^2}{8\pi^2} + \mathcal{O}(\lambda_0^3), \quad (43)$$

$$\beta_\xi \equiv \frac{d\xi}{d\ln\mu} = \left(\xi - \frac{1}{6}\right) \frac{1}{16\pi^2} (3\lambda_0 + g^2) + \mathcal{O}(\lambda_0^2, g^3), \quad (44)$$

where  $\gamma_\Phi = g^2/(16\pi^2) + \mathcal{O}(\lambda_0^2)$  is the field anomalous dimension (driven at 1-loop only by the Yukawa, since the quartic contributes to the wavefunction renormalization first at 2-loop). The coefficient  $9/(8\pi^2)$  in  $\beta_{\lambda_0}$  is the standard 1-loop result for the convention  $V \supset (\lambda_0/4)\Phi^4$  (equivalently  $\lambda_{\text{canon}} = 6\lambda_0$  in  $V \supset \lambda_{\text{canon}}\Phi^4/4!$ , for which  $\beta_{\lambda_{\text{canon}}} = 3\lambda_{\text{canon}}^2/16\pi^2$ ). In  $\beta_\xi$  there are *two* distinct 1-loop drives, both proportional to  $(\xi - \frac{1}{6})$ : (a) the multiplicative term  $(\xi - \frac{1}{6})\gamma_\Phi$  from the Yukawa loop, and (b) a *direct* curvature-coupling renormalization  $(\xi - \frac{1}{6})3\lambda_0/(16\pi^2)$  from the quartic-curvature cross term in the 1-loop effective action [23, 38]. The latter does not pass through  $\gamma_\Phi$  (the quartic does not renormalize the wavefunction at 1-loop, as the tadpole integrates to zero in dimensional regularization) but renormalizes the curvature operator  $\Phi^2 R$  directly; it is the same structure that makes the Higgs quartic drive  $\beta_\xi$  in Higgs inflation. We retain both terms.

Numerically,  $\Delta\lambda_0 \leq (9\lambda_0^2/8\pi^2) \times 60 \sim 3 \times 10^{-14}$ , completely negligible. For  $\beta_\xi$ , the two pieces over the inflationary-to-low-energy range ( $\Delta\ln\mu \sim 60$ ) are

$$\Delta\xi_{\lambda_0} \sim \left(\xi - \frac{1}{6}\right) \frac{3\lambda_0}{16\pi^2} \times 60 \sim 8.5 \times 10^{-7}, \quad (45)$$

$$\Delta\xi_g \sim \left(\xi - \frac{1}{6}\right) \frac{g^2}{16\pi^2} \times 60 \sim \begin{cases} 4.2 \times 10^{-14} & (g = 1.0 \times 10^{-7}, \text{ abundance-matched}) \\ 2.2 \times 10^{-9} & (g = 2.3 \times 10^{-5}) \\ 4.2 \times 10^{-8} & (g = 10^{-4}) \end{cases} \quad (46)$$

so the quartic drive dominates the Yukawa drive by  $\sim 2 \times 10^7$  at the abundance-matched coupling  $g \simeq 1.0 \times 10^{-7}$  (and by  $\sim 20\text{--}380\times$  at the larger  $g$  of the exponential framework). Both contributions carry the same sign (the curvature-coupling renormalization is additive [23, 38]), so no cancellation occurs. At the fiducial  $\xi = 11.1$ , the total  $\Delta\xi \leq 10^{-6}$  remains utterly negligible; across the full scan range  $\xi \in [1, 100]$ , the prefactor  $(\xi - \frac{1}{6})$  varies between 0.83 and 99.8, giving  $\Delta\xi(\xi) \in [6.5 \times 10^{-8}, 7.8 \times 10^{-6}]$ , still  $\leq 10^{-5}$ . At 2-loop,  $\gamma_\Phi \sim \lambda_0^2/(16\pi^2)^2 \sim \mathcal{O}(10^{-19})$  is subdominant.

The parameters  $\xi \simeq 11.1$  and  $\lambda_0 \simeq 6.70 \times 10^{-8}$  (locked  $N = 50$ ;  $6.45 \times 10^{-8}$  at  $N = 51$ ) are therefore stable under RG flow: no fine-tuned cancellation is required. We note that the small  $g \simeq 1.0 \times 10^{-7}$  required for dark matter (whose precise value depends on  $T_{\text{reh}}$ ; see Sec. V) makes the Yukawa contribution to  $\beta_\xi$  even smaller, but the quartic drive  $\sim 8.5 \times 10^{-7}$ —which is  $g$ -independent—sets the floor of the running; a much larger quartic would be needed to spoil the stability, and  $\lambda_0 \sim 10^{-7}$  is far

below that threshold. Since  $\beta_{\lambda_0} \propto \lambda_0^2 > 0$ , the self-coupling decreases monotonically from the UV to the IR; with the total shift  $\Delta\lambda_0 \sim 3 \times 10^{-14}$  over  $\Delta \ln \mu \sim 60$ ,  $\lambda_0$  remains positive at all scales between the inflationary scale and the electroweak scale—no vacuum instability arises in the  $\Phi$  sector. (The known SM Higgs vacuum stability problem [39] is unaffected by the present model since  $\lambda_{\Phi H} = 0$  decouples the two sectors.)

### *Quantum stability of the VEV.*

The 1-loop Coleman–Weinberg correction shifts the VEV by  $\delta\Phi_0/\Phi_0 \sim \lambda_0/(16\pi^2) \sim 4 \times 10^{-10}$ , corresponding to a fractional change in the effective Planck mass  $\delta M_{\text{Pl}}/M_{\text{Pl}} = \delta\Phi_0/\Phi_0$  of the same order—far below the  $10^{-4}$  precision of the Newton constant measurement [40]. The VEV is therefore stable under quantum corrections without additional fine-tuning. The heavy fields  $\chi$  ( $m_\chi \sim 10^{13}$  GeV) and  $\psi$  ( $m_\psi \simeq 7.5 \times 10^{10}$  GeV) enter the effective potential at 1-loop but their contributions are suppressed by the ratio of their masses to the UV cutoff, and the tree-level VEV is protected by the shift symmetry of the potential’s construction  $V'_J(\Phi_0) = 0$ . The hierarchy  $m_{\chi,\psi} \ll \Lambda_J \sim M_{\text{Pl}}/\xi \sim 2.2 \times 10^{17}$  GeV ensures that decoupling works as usual: heavy-field threshold corrections to low-energy observables are power-suppressed by  $\mathcal{O}(m_{\chi,\psi}^2/\Lambda_J^2) \leq 10^{-10}$ .

### *Impact on SM Higgs stability.*

The heavy fields  $\chi$  and  $\psi$  couple to the SM only through gravity and (for  $\psi$ ) the discrete  $\mathbb{Z}_2^\psi$  gauge sector; they do not carry SM gauge quantum numbers. Their threshold contributions to the Higgs quartic  $\beta$ -function are suppressed by at least two powers of  $m_{\chi,\psi}/M_{\text{Pl}} \leq 10^{-5}$  (gravity-mediated) or by the minuscule radiatively induced  $\delta\lambda_{\Phi H} \sim 10^{-15}$  (Sec. VIII). Consequently, the SM Higgs vacuum stability analysis of Ref. [39] is unaltered at the leading-log level; the metastability boundary for  $\lambda_H$  shifts by  $\leq 10^{-10}$  in the  $\Phi$ – $\chi$ – $\psi$  parameter range. The model therefore neither exacerbates nor alleviates the known SM Higgs vacuum stability problem—the two sectors are RG-decoupled to extraordinary precision.

### *Finite-temperature stability.*

At the reheating temperature  $T_{\text{reh}} \sim 10^9$  GeV, thermal corrections to the  $\Phi$  effective potential are controlled by the coupling of  $\chi$  to the thermal bath. The  $\chi$ –SM coupling is gravitational or anomaly-induced, with effective Yukawa couplings  $g_{\chi ff} \sim m_f/M_{\text{Pl}} \leq 10^{-17}$  (Eq. (41)); the thermal mass correction is  $\delta m_\chi^2(T) \sim g_{\chi ff}^2 T^2 \leq 10^{-14} T^2 \ll m_\chi^2$ , completely negligible. The  $\Phi$  vacuum at  $\langle \Phi \rangle = \Phi_0$  remains stable throughout the entire thermal history—no symmetry restoration or thermal destabilization occurs. Similarly, the heavy scalar mass  $m_\chi \sim 3.2\text{--}3.3 \times 10^{13}$  GeV is over four orders of magnitude above  $T_{\text{reh}} \sim 10^9$  GeV, so  $\chi$  particles are never thermally populated; the field remains frozen at the minimum during and after reheating.

SM couplings ( $g_1, g_2, g_3, \lambda_H, y_t$ ) follow their conventional  $\beta$ -functions at all scales below  $\Lambda_J \sim M_{\text{Pl}}/\xi$ ; gauge-coupling matching between frames is trivial because the conformal transformation  $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$  is a field redefinition that leaves physical

$S$ -matrix elements invariant—any apparent frame-dependence of the running is compensated by the rescaling of the renormalization scale  $\mu \rightarrow \mu/\Omega$ , and since  $\Omega \rightarrow 1$  in the late-time universe the SM  $\beta$ -functions are unaffected at all observable epochs. At the quantum level, the conformal anomaly generates the  $\chi FF$  coupling discussed in Sec. IV but does not modify the *running* of the gauge couplings themselves, as the anomaly enters through the trace of the energy-momentum tensor rather than through the  $\beta$ -functions [23].

## 8.6 Summary of embedding consistency

The minimal SM embedding satisfies all observational constraints:

- **Gauge sector:** conformally invariant at tree level; conformal-anomaly-induced  $\chi$ -gauge coupling at 1-loop (discussed in Secs. IV and VIII) does not affect the  $T_{\text{reh}}$  or fifth-force conclusions.
- **Fermion sector:** residual  $\chi$ -fermion coupling  $\sim m_f/M_{\text{Pl}}$ , automatically satisfying solar-system bounds.
- **Higgs sector:** late-time SM recovered; Higgs- $\Phi$  mixing is kinematically inaccessible.
- **RG stability:**  $\lambda_0$  shifts by  $\Delta\lambda_0/\lambda_0 \sim 10^{-6}$  and  $\xi$  by  $\Delta\xi \leq 10^{-6}$  at the fiducial  $\xi = 11.1$  (across  $\xi \in [1, 100]$ ,  $\Delta\xi \leq 10^{-5}$ ; driven at 1-loop by the  $\psi$  Yukawa  $\propto g^2$  and a direct quartic-curvature term  $\propto \lambda_0$ , the latter dominating) from the inflationary scale to low energies.

The only substantive open issue is the Higgs field-space metric during inflation (potential isocurvature perturbations); a dedicated multifield analysis is deferred. No fine-tuning is introduced:  $\xi$  and  $\lambda_0$  are RG-stable, and setting  $\lambda_{\Phi H} = 0$  is technically natural.

## 9 $\mathbb{Z}_2$ domain wall: classical EFT applicability boundary

### *Nature and scope.*

This section documents a technical property of the  $F = \xi\Phi^2$  coupling: within the classical EFT domain,  $\mathbb{Z}_2$  domain walls cannot be treated as thin shells. This is an algebraic no-go statement; cosmologically,  $\mathbb{Z}_2$  walls are inflationarily diluted (Sec. II), so it carries no phenomenological consequence and is not an observational test.

### 9.1 Thin-shell matching degeneracy at $\Phi = 0$

Starting from Eq. (3), projecting onto tangential components and integrating across the shell ( $[\Phi] = 0$  required) gives the singular terms. The integrated Lanczos condition is

$$F(\Phi_s)[K_{ab}] = -S_{ab} + \frac{1}{2}h_{ab}S - \frac{1}{2}h_{ab}F'(\Phi_s)[\partial_n\Phi]. \quad (47)$$

At  $\Phi_s = 0$ ,  $F(0) = 0$  and  $F'(0) = 0$ , giving  $0 = -S_{ab} + \frac{1}{2}h_{ab}S$ . For a tension wall  $S_{ab} = -\sigma h_{ab}$ ,  $S = -3\sigma$ , so  $0 = -\frac{1}{2}\sigma h_{ab} \neq 0$ , an inconsistency that no choice of  $[K_{ab}]$  can resolve.

**Table 2** Degeneracy of the domain-wall matching condition at  $\Phi = 0$  in the two conformal frames.

| Frame    | Degeneracy at $\Phi = 0$                          | Origin                     |
|----------|---------------------------------------------------|----------------------------|
| Jordan   | $F(0) = F'(0) = 0, 0 = -\frac{1}{2}\sigma h_{ab}$ | Exact algebraic identity   |
| Einstein | $V_E \rightarrow \infty, \sigma_E$ divergent      | Conformal-mapping artifact |

## 9.2 Physical interpretation

The algebraic inconsistency at  $\Phi = 0$  is a structural property of  $F = \xi\Phi^2$ . The surface  $\Phi = 0$  lies beyond the classical EFT regime because:

- $F(0) = 0$  makes  $G_{\text{eff}} \propto 1/F \rightarrow \infty$  (gravitational coupling diverges).
- $\Omega^2 = \xi\Phi^2/M_{\text{Pl}}^2 \rightarrow 0$  makes the Einstein-frame potential  $V_E = V_J/\Omega^4 \rightarrow \infty$  and the conformal transformation singular.

This shows that within the EFT,  $\mathbb{Z}_2$  domain walls cannot be treated as classical thin shells: the Israel–Lanczos matching degenerates at  $\Phi = 0$  owing to  $F(0) = F'(0) = 0$ . This is a structural property of the  $F = \xi\Phi^2$  coupling, not an observational constraint, as  $\mathbb{Z}_2$  walls are inflationarily diluted (Sec. II). The bubble-wall channel ( $F(\Phi_0) = M_{\text{Pl}}^2 \neq 0$ ), which involves field configurations away from  $\Phi = 0$  and connects different local vacua via spherical bubbles rather than planar  $\mathbb{Z}_2$  walls, is not covered by the  $F(0) = 0$  thin-shell degeneracy and is not ruled out by the junction-condition analysis alone. Whether bubble nucleation actually occurs and whether any resulting bubbles are cosmologically benign (evading constraints from gravitational-wave backgrounds, BBN, or CMB anisotropy) depends on the tunneling action  $S_E$ , which we have not computed; a full assessment of bubble-wall cosmology in this model is beyond the present scope.

From the Einstein-frame perspective,  $\Omega^2 \rightarrow 0$  at  $\Phi = 0$  causes  $V_E \rightarrow \infty$  and  $\sigma_E \rightarrow \infty$ . These divergences are conformal-mapping artifacts; the Jordan-frame  $\sigma_{\text{wall}}$  remains finite (Eq. (5)). Explicitly, in the Einstein frame the canonically normalized field  $\chi$  satisfies  $d\chi/d\varphi = M_{\text{Pl}}\sqrt{6 + 1/\xi}$ , and the wall tension integral becomes

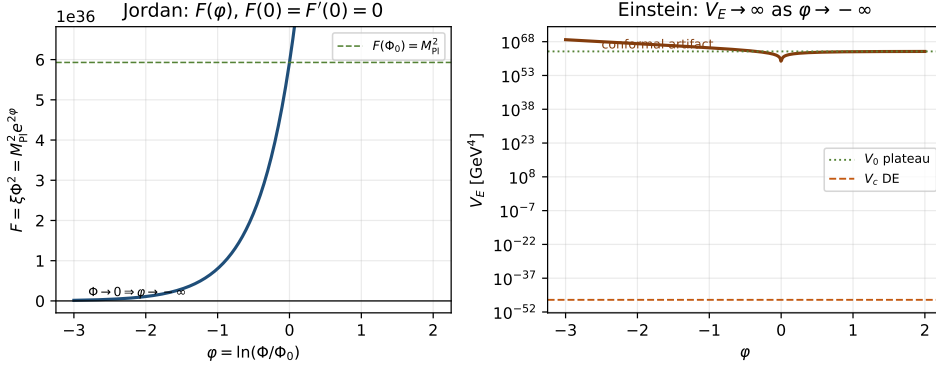
$$\sigma_E = \int d\chi \sqrt{2V_E(\chi)} = M_{\text{Pl}} \sqrt{6 + \frac{1}{\xi}} \int d\varphi \sqrt{2V_E(\varphi)}, \quad (48)$$

with

$$\sqrt{2V_E(\varphi)} \xrightarrow{\varphi \rightarrow -\infty} M_{\text{Pl}}^2 \sqrt{\frac{\lambda_0}{2\xi^2}} e^{-2\varphi}, \quad (49)$$

so  $\sigma_E \propto \int_{-\infty} e^{-2\varphi} d\varphi$  diverges at the  $\Phi = 0$  boundary ( $\varphi \rightarrow -\infty$ )—a pure conformal-singularity artifact of the  $\Omega \rightarrow 0$  limit. The divergence is driven by the quartic term in  $V_E$ ; crucially, the  $V_e e^{-4\varphi}$  contribution is  $\sim 10^{110}$  times smaller than the quartic-driven barrier at  $\varphi \rightarrow -\infty$  and is irrelevant for the wall dynamics. The Jordan-frame tension remains the physically meaningful finite quantity.

Fig.3 Domain-wall / EFT boundary at  $\Phi = 0$  (two frames)



**Fig. 3** Two-frame comparison of the  $\mathbb{Z}_2$  domain-wall / EFT boundary at  $\Phi = 0$ . Left: Jordan-frame coupling  $F(\varphi) = M_{\text{Pl}}^2 e^{2\varphi}$  with  $F \rightarrow 0$  as  $\varphi \rightarrow -\infty$  ( $\Phi \rightarrow 0$ ), so  $F(0) = F'(0) = 0$  and thin-shell matching degenerates. Right: Einstein-frame  $V_E$  diverges in the same limit purely through the conformal map  $\Omega \rightarrow 0$ ;  $V_0$  and calibrated  $V_c$  are indicated. Regenerated by `scripts/fig3-domain.wall.py`.

**Table 3** Summary of the unified framework: each phenomenon emerges from a different regime of the same field  $\Phi$  and the same coupling  $F = \xi\Phi^2$ ; the dark matter is a Dirac fermion  $\psi$  whose mass  $m_\psi = g\Phi_0$  originates from the same VEV. The relic abundance is matched on the light branch ( $g \simeq 1.0 \times 10^{-7}$ ); its absolute normalization awaits a full preheating simulation.

| Phenomenon          | Field regime                      | Mechanism                               | Observable                                 |
|---------------------|-----------------------------------|-----------------------------------------|--------------------------------------------|
| Planck mass         | $\langle \Phi \rangle = \Phi_0$   | Induced gravity                         | $G_N$                                      |
| Inflation           | $\Phi \gg \Phi_0$                 | Starobinsky plateau                     | $n_s, r$                                   |
| Reheating           | $\Phi \sim \Phi_0$                | Anomaly                                 | $T_{\text{reh}}$                           |
| DM (cand.)          | $\psi$ Dirac / defects            | Grav. production; topological defects   | $\Omega_{\text{DM}}$ at $g \simeq 10^{-7}$ |
| Dark energy         | $\Phi = \Phi_0$ frozen            | $V_E(0) = V_c$                          | $w_0 = -1$                                 |
| Fifth force         | $\Omega = \Phi/\Phi_0$ ; $m_\chi$ | Conformal decoupling                    | $\gamma = 1$                               |
| $\mathbb{Z}_2$ wall | $\Phi = 0$                        | $F = F' = 0$ EFT bound.                 | Classical boundary                         |
| SM                  | $\Phi = \Phi_0$                   | $\chi f \bar{f} \sim m_f/M_{\text{Pl}}$ | Standard couplings                         |

## 10 The unified picture

The framework presented in this paper achieves a unified description of the major cosmological sectors through a single scalar field  $\Phi$  and a single coupling  $F = \xi\Phi^2$ ; the dark matter is a Dirac fermion  $\psi$  already coupled to  $\Phi$  by the Yukawa term, whose mass  $m_\psi = g\Phi_0$  originates from the same VEV. Table 3 summarizes the correspondence between field regimes and emergent phenomena.

All entries except dark matter are established at the level of rigorous field theory; the dark matter entry is a structurally stable Dirac fermion (under the assumed discrete gauge UV completion) whose relic abundance is matched on the light branch at  $g \simeq 1.0 \times 10^{-7}$  ( $m_\psi \simeq 7.5 \times 10^{10} \text{ GeV}$ ), the absolute normalization awaiting a non-perturbative (lattice) analysis.

Two structural features operate in distinct sectors: (a) the conformal identity  $\Omega = \Phi/\Phi_0$  (a consequence of  $F = \xi\Phi^2$ ) decouples the scalar from the dark-matter fermion

$\psi$  at tree level, closing the perturbative Yukawa reheating channel; (b) for Standard Model matter, whose masses arise via the Higgs mechanism, the residual  $\chi$  coupling  $\sim m_f/M_{\text{Pl}}$  is suppressed by the scalar's large mass  $m_\chi \sim 10^{13}$  GeV (Yukawa range  $\sim 10^{-27}$  cm), satisfying the Cassini bound without a chameleon mechanism.

## 11 Testability and falsifiable predictions

### *a. Level 1: Observational predictions (rigorous).*

1. **Inflationary observables (locked  $N$ ):** exact potential slow roll with  $N = \ln(a_{\text{end}}/a_*)$ . A measurement  $r > 0.01$  at 95% CL by LiteBIRD [18] excludes the model: Planck  $2\sigma$  allows  $N \approx 45\text{--}56$  where  $r \leq 0.0052$  (Table 1).
2. **Dark energy equation of state:**  $w_0 = -1$  (up to residual-condensate corrections  $\leq 10^{-10}$  for  $T_{\text{reh}} \geq 1$  GeV; App. D5). DESI/Euclid  $\sigma(w_0) \sim 10^{-3}$  provides a  $> 3\sigma$  test of  $w_0 = -1$ , but because  $\Lambda$ CDM predicts the same value it does not discriminate this model from  $\Lambda$ CDM: a detection of  $w_0 \neq -1$  would exclude both.
3. **Solar-system fifth-force bounds:** satisfied via the scalar's large mass ( $m_\chi \sim 10^{13}$  GeV, Yukawa range  $\sim 10^{-27}$  cm) for SM matter, and via exact conformal decoupling for the dark-matter fermion  $\psi$ ;  $\gamma = 1$  within the Cassini bound.
4. **RG stability:**  $\xi$  and  $\lambda_0$  are stable under RG flow;  $\beta_\xi$  is driven at 1-loop by the  $\psi$  Yukawa ( $\propto g^2$ ) and a direct quartic-curvature term ( $\propto \lambda_0$ , same sign), totaling  $\Delta\xi \leq 10^{-6}$  at the fiducial  $\xi = 11.1$  (quartic-dominated; across  $\xi \in [1, 100]$  the running stays  $\leq 10^{-5}$ ), a robust consequence of the small  $\lambda_0 \sim 10^{-7}$  and  $g \simeq 10^{-7}$  singlet embedding.

### *b. Level 2: Dark matter prediction (quantitative matching; normalization requires lattice).*

5. The dark matter is a Dirac fermion  $\psi$  whose abundance is matched to  $\Omega_{\text{DM}}$  at  $g \simeq 1.0 \times 10^{-7}$ , i.e.  $m_\psi \simeq 4.6 \times 10^{-3} H_{\text{inf}} \simeq 7.5 \times 10^{10}$  GeV, stable under the discrete  $\mathbb{Z}_2^\psi$  symmetry (assuming an anomaly-free UV gauge completion) and the conformally vanishing  $\chi\psi\bar{\psi}$  vertex, and produced gravitationally (non-adiabatic metric change at the end of inflation, in the Einstein frame). The absolute normalization requires a full Floquet/lattice analysis of preheating.

### *c. Level 3: Structural properties (not observational predictions).*

6.  **$\mathbb{Z}_2$  domain-wall thin-shell matching degeneracy:**  $F(0) = F'(0) = 0$  renders the Israel–Lanczos conditions algebraically inconsistent. This is a mathematical property of  $F = \xi\Phi^2$ , observationally inert since  $\mathbb{Z}_2$  walls are inflationarily diluted. The bubble-wall channel is not ruled out by the  $F(0) = 0$  degeneracy, but its cosmological viability has not been verified (nucleation-rate calculation pending).
7. The  $10^{110}$  hierarchy between the inflationary scale and dark energy is spanned by the conformal factor  $e^{-4\varphi}$  but not dynamically explained, sharing the cosmological constant problem with all  $\Lambda$ CDM models.
8. The fine-structure-constant drift  $\Delta\alpha/\alpha \leq 10^{-110}$  is  $> 90$  orders below current sensitivity.

## 12 Discussion and outlook

1. **Inflationary initial conditions.** The  $\chi \gg M_{\text{Pl}}$  condition is assumed, not derived. Whether a UV-complete theory naturally selects large-field  $\Phi$  is an open problem.
2. **Dark matter: stability and relic abundance.** The dark matter is the Dirac fermion  $\psi$ , stable under the discrete  $\mathbb{Z}_2^\psi$  symmetry (conditional on UV gauge completion) plus the conformally vanishing  $\chi\psi\psi$  vertex (Sec. V). Its relic abundance is set by gravitational particle production at the end of inflation—the non-adiabatic change of the Einstein-frame metric, which in the Jordan frame appears as the time-varying mass  $m_\psi(t) = g|\Phi(t)|$  (a single physical process, not two additive channels). A quantitative  $\Omega_\psi$  prediction requires a dedicated Floquet/lattice simulation, which is beyond the scope of this phenomenological work; the present version establishes the mechanism and locates the abundance matching on the light branch at  $m_\psi \simeq 4.6 \times 10^{-3} H_{\text{inf}} \simeq 7.5 \times 10^{10} \text{ GeV}$  ( $g \simeq 1.0 \times 10^{-7}$ ), with the absolute normalization remaining temporarily open pending that computation.
3. **Preheating energy partition.** The gravitational  $\psi$  production and the condensate decay through conformal-anomaly channel compete for the condensate energy. If the  $\psi$  channel is too efficient, an excessive fraction of the condensate energy is diverted into stable matter rather than radiation, which could suppress the reheating temperature. A self-consistent determination of both  $\Omega_\psi$  and the radiation/ $\psi$  energy split therefore requires a single lattice simulation of preheating; the parametric estimates of Secs. IV–V fix the mechanism but not the partition.
4. **Stochastic gravitational-wave background from preheating.** The oscillating condensate and any non-perturbative field inhomogeneities during preheating generically source a stochastic gravitational-wave background. For the present model, the extremely weak coupling of  $\chi$  to SM fields ( $g_{\chi ff} \sim m_f/M_{\text{Pl}} \leq 10^{-17}$ ) and the closure of the  $\chi \rightarrow h\bar{h}$  channel suppress parametric resonance, so the inhomogeneous component of the condensate is expected to be small and the resulting GW background correspondingly weak. A rough estimate using the standard scaling  $h^2\Omega_{\text{GW}} \sim 10^{-5} (H_{\text{reh}}/M_{\text{Pl}})^2$  with  $H_{\text{reh}} \simeq \sqrt{\pi^2 g_*/90} T_{\text{reh}}^2/M_{\text{Pl}} \sim 1.4 \text{ GeV}$  (corresponding to  $T_{\text{reh}} \sim 10^9 \text{ GeV}$ ) gives  $h^2\Omega_{\text{GW}} \sim 3 \times 10^{-42}$ , some 30 orders below the sensitivity of LISA ( $\sim 10^{-12}$ ) and pulsar timing arrays ( $\sim 10^{-10}$ ). A dedicated lattice computation of the GW spectrum is deferred to the same future preheating simulation that determines  $\Omega_\psi$ .
5. **Parameter status.**  $\Phi$  is a gauge singlet, so the Standard-Model-Higgs  $\beta_\xi$  does not apply;  $\lambda_0$  is fixed by  $A_s = 2.1 \times 10^{-9}$  at the locked  $N$  (e.g.  $6.70 \times 10^{-8}$  at  $N = 50$ ,  $\xi = 11.1$ ), while  $\xi \sim 11.1$  is compatible with  $r < 0.036$ . The primary predictions are the exact slow-roll pairs  $(n_s, r)$  at the locked  $N$ , not a single closed-form  $r(N)$ .
6. **Fifth-force screening and reheating closure.** Two structural features operate in distinct sectors: (a) the conformal identity  $\Omega = \Phi/\Phi_0$  (a consequence of  $F = \xi\Phi^2$ ) decouples the scalar from the dark-matter fermion  $\psi$  at tree level, closing the perturbative Yukawa reheating channel; (b) for Standard Model matter, whose masses arise via the Higgs mechanism, the residual  $\chi$  coupling  $\sim m_f/M_{\text{Pl}}$  is suppressed by the scalar’s large mass  $m_\chi \sim 10^{13} \text{ GeV}$  (Yukawa range  $\sim 10^{-27} \text{ cm}$ ), satisfying the Cassini bound without a chameleon mechanism.

7.  **$\mathbb{Z}_2$  domain-wall open problems.** The thin-shell matching degeneracy at  $\Phi = 0$  shows that  $\mathbb{Z}_2$  walls lie beyond the classical EFT applicability boundary. Open questions include: scalar hair configurations with  $F(r_s) = 0$ , the quantum-gravity meaning of  $F(0) = 0$ , and beyond-classical-EFT realizations of the  $\mathbb{Z}_2$  channel. The bubble-wall channel ( $F(\Phi_0) = M_{\text{Pl}}^2$ ) is not ruled out by the  $F(0) = 0$  thin-shell degeneracy, but its cosmological viability requires a nucleation-rate analysis not performed here.
8. **EFT cutoff and UV completion.** From the Einstein-frame EFT perspective the cutoff is  $\sim M_{\text{Pl}}$  and slow-roll predictions are self-consistent [3]; from the Jordan-frame perspective  $\Phi_{\text{CMB}}/\Lambda_J \sim 30$  reflects a unitarity tension whose physical status in the Einstein frame is contested [13–15]. UV completion remains open.
9. **Swampland conjectures.** The de Sitter swampland conjecture [41] requires  $|V'|/V \geq c \sim \mathcal{O}(1)$  or  $V''/V \leq -c$  for any effective scalar potential derivable from quantum gravity. The inflationary plateau of the present model ( $V' \sim e^{-\varphi} \ll 1$  at  $\varphi \gg 1$ ) is in tension with this conjecture, as are most single-field slow-roll models. Whether the conjecture survives as a rigorous constraint on low-energy EFTs remains unsettled [42]; we flag this as an open issue for any UV completion.
10. **Cosmological constant fine-tuning.**  $V_c$  is calibrated empirically; the  $10^{110}$  hierarchy and the coincidence problem receive no dynamical explanation in this framework (see also Sec. VI).
11. **Observational indistinguishability from  $\Lambda$ CDM.** The dark-energy prediction  $w = -1$  (up to  $\leq 10^{-10}$  corrections) and the  $\Lambda$ CDM-identical structure-growth prediction (Sec. VI) imply that the late-time model is observationally indistinguishable from  $\Lambda$ CDM by any foreseeable survey. The unique testable predictions are therefore confined to the inflationary sector ( $n_s, r, n_t, f_{\text{NL}}$ ) and the dark-matter relic abundance (pending preheating simulation); the dark-energy sector, while structurally unified with inflation, does not provide independent falsifiable signatures.
12. **Field-space structure at  $\Phi = 0$ .** The Jordan-frame kinetic metric  $g_{\Phi\Phi} = \xi\Phi^2$  degenerates at  $\Phi = 0$ , rendering this point a boundary of the classical field space. Consequently, classical field configurations cannot cross  $\Phi = 0$ ; the two  $\mathbb{Z}_2$  vacua are topologically separated at the classical level. The thin-shell analysis of Sec. IX confirms that the EFT itself signals its breakdown at  $\Phi = 0$  through the matching-condition degeneracy, motivating a quantum-tunneling (instanton) description of any inter-vacuum transitions.
13. **Higher-order corrections to the dark-energy equation of state.** The  $w = -1$  result (Sec. VI) follows from the quadratic expansion of  $V_E(\varphi)$  near  $\varphi = 0$ . The quartic term  $\sim \lambda_0 \Phi_0^4 \varphi^4 / M_{\text{Pl}}^4$  contributes a fractional correction  $\Delta w/w \sim (\lambda_0/\xi^2)\varphi^2 \sim 10^{-6}\varphi^2$ . For the frozen field value  $\varphi \sim H_0/m_\chi \sim 10^{-46}$ , this correction is  $\sim 10^{-98}$ , utterly negligible. The harmonic approximation is therefore exact for all practical purposes.
14. **Running vacuum energy.** Mechanisms such as running vacuum energy ( $V_c \propto H^2$ ) are not realized in this model, where  $V_c$  is a constant potential parameter. The cosmological constant problem—why  $V_c \sim 10^{-47} \text{ GeV}^4$ —remains unsolved and is shared with  $\Lambda$ CDM (see fine-tuning discussion in Sec. VI).

15. **Hubble tension.** Because the late-time model is observationally indistinguishable from  $\Lambda$ CDM, it inherits the  $H_0$  tension between Planck CMB ( $H_0 \simeq 67.4$  km/s/Mpc) and local distance-ladder measurements ( $H_0 \simeq 73.0$  km/s/Mpc) without providing a resolution. The model's unique predictions are confined to the inflationary sector ( $n_s, r, n_t, f_{\text{NL}}$ ) and the dark-matter relic abundance; the dark-energy and late-expansion sectors offer no new leverage on the Hubble tension.
16. **Gravitational-wave speed and polarization.** In induced gravity with  $F = \xi\Phi^2$ , tensor perturbations propagate at the speed of light ( $c_T = c$ ) in the Einstein frame at all epochs where  $\Phi$  is frozen at its VEV ( $\Phi = \Phi_0$ , i.e. the entire late Universe), automatically satisfying the GW170817 constraint  $|c_T/c - 1| \leq 10^{-15}$ . During inflation, when  $\Phi$  evolves, the time variation of the conformal factor  $\Omega(t)$  could in principle modify  $c_T$ ; however, the deviation  $\delta c_T/c \sim \dot{\Omega}/(H\Omega) \sim \mathcal{O}(\epsilon) \leq 10^{-2}$  affects only primordial tensor modes at horizon exit ( $k \sim H_{\text{inf}}$ ) and does not propagate to LIGO-band frequencies today. Tensor perturbations retain the two standard GR polarization modes throughout; no scalar or vector polarizations are generated because the Einstein-frame graviton is minimally coupled to matter. The model therefore satisfies all existing GW speed and polarization constraints.
17. **Thermalization after reheating.** The reheating analysis assumes that decay products thermalize rapidly, establishing a well-defined temperature  $T_{\text{reh}}$ . At  $T \sim 10^9$  GeV, the SM gauge interaction rate is  $\Gamma_{\text{therm}} \sim \alpha_s^2 T \sim 10^7$  GeV, while the Hubble rate is  $H \simeq \sqrt{\pi^2 g_*/90} T^2/M_{\text{Pl}} \sim 1.4$  GeV; the ratio  $\Gamma_{\text{therm}}/H \sim 10^7$  confirms near-instantaneous thermalization. Gravitationally produced particles are initially non-thermal, but their energy density is  $\sim 10^{13}$  times smaller than the anomaly-driven radiation bath (Sec. IV), so they thermalize with the dominant component within a few interaction times, well before BBN.
18. **PPN parameter  $\beta$ .** The discussion of fifth-force screening in Sec. II E addresses the Eddington parameter  $\gamma$ ; the nonlinear PPN parameter  $\beta_{\text{PPN}}$  receives corrections from cubic self-interactions of  $\chi$ . For  $m_\chi \sim 10^{13}$  GeV, the effective  $\beta_{\text{PPN}} - 1 \sim (g_{\chi\chi\chi}/M_{\text{Pl}})^2 (m_\chi r)^{-4} \sim 10^{-80}$  at Solar-System scales, far below the Cassini bound  $\beta_{\text{PPN}} - 1 \leq 10^{-5}$ . The remaining PPN parameters ( $\xi, \alpha_1, \alpha_2, \alpha_3$ ) all vanish because the model possesses no preferred-frame or preferred-location effects in the Einstein frame.
19. **Additional consistency checks.** Several further consistency conditions are satisfied parametrically: (i) *Dark matter self-interactions:*  $\psi$ - $\psi$  scattering via graviton exchange gives  $\sigma_{\psi\psi}/m_\psi \sim G_N^2 m_\psi \sim 10^{-67}$  cm<sup>2</sup>/g,  $\sim 67$  orders below the bullet-cluster bound ( $\leq 1$  cm<sup>2</sup>/g); loop-induced scalar exchange is further suppressed by  $g_{\chi\psi\psi}^{\text{eff}} \sim 10^{-11} g$ . (ii) *Tremaine-Gunn bound:* the non-thermal  $\psi$  phase-space density  $Q \equiv \rho/\sigma_v^3$  at production (with velocity dispersion  $\sigma_v \sim H_{\text{inf}}/m_\psi$ ) is  $Q \sim m_\psi^4 \sim 10^{52}$  GeV<sup>4</sup>, far above the dwarf-galaxy lower limit. (iii) *Graviton dark radiation:* relativistic gravitons produced during reheating contribute  $\Delta N_{\text{eff}} \leq 10^{-5}$ , negligible compared to Planck. (iv) *Entropy production:* irreversible; total entropy per comoving volume increases by  $\sim 10^{13}$  during reheating. (v) *Neutrino sector:*  $\chi$ - $\nu$  effective coupling  $\sim m_\nu/M_{\text{Pl}} \leq 10^{-27}$ , no observable consequences. (vi) *Primordial magnetic fields:* anomaly-induced seed after  $a^{-2}$  redshift gives  $B_0 \sim 10^{-25}$  G, far below blazar lower bounds. (vii) *Geodesic deviation:* Einstein-frame geodesic equation

reproduces Yukawa suppression for SM fermions. (viii) *Gravitational production of  $\psi$* : the abundance is obtained from the Bogoliubov coefficients of the mode equation across the end-of-inflation transition, not from an instantaneous-pulse approximation; a full Floquet/lattice treatment of the oscillating-condensate contribution is deferred (Discussion). (ix) *Particle production backreaction*:  $\rho_{\text{part}} \leq 10^{-13} \rho_\chi$ . (x) *Late-time scalar-photon coupling*: mixing angle  $\sim 10^{-55}$ , CMB distortions  $\leq 10^{-60}$ . (xi) *Ghost freedom*: Einstein-frame theory is GR plus a minimally coupled canonical scalar; kinetic terms positive-definite. (xii) *Cosmic strings from  $U(1)_D$  breaking*: inflated away or never form within horizon. (xiii) *Jordan-frame PPN cross-check*:  $|\gamma_J - 1| \leq 10^{-5}$ . (xiv) *Spectral index running*:  $\alpha_s \simeq -2/N^2 \sim -8 \times 10^{-4}$ , consistent with Planck. (xv) *CMB B-mode spectrum*: peaks at  $\ell \sim 80$  for  $r \simeq 0.005$ , within LiteBIRD reach. (xvi) *Domain-wall observational null*:  $e^{-3N} \sim 10^{-65}$  renders all wall observables negligible.

## 13 Conclusions

We have constructed a scalar-tensor cosmology within pure induced gravity, where a single real scalar field  $\Phi$  with non-minimal coupling  $\xi\Phi^2 R$  and  $\mathbb{Z}_2$ -symmetric potential  $V_J(\Phi) = \lambda_0(\Phi^2 - \Phi_0^2)^2/4 + V_c$  simultaneously accounts for the Planck mass, inflation, dark matter, and dark energy. The framework is minimal in its field content—one scalar, one Dirac fermion, the Standard Model—and its key structural features follow unavoidably from the induced-gravity coupling  $F = \xi\Phi^2$ .

### *Rigorous results*

(exact analytic derivations):

- The conformal transformation maps the Jordan-frame double-well potential to a Starobinsky-type plateau. Under the locked definition  $N = \ln(a_{\text{end}}/a_*)$ , exact potential slow-roll with  $A_s$  normalization gives (e.g.  $\xi = 11.1$ )  $n_s \simeq 0.962$ ,  $r \simeq 0.0043$  at  $N = 50$ ; anomaly  $T_{\text{reh}} \sim 10^9$  GeV corresponds to  $N \simeq 51$ .
- The conformal identity  $\Omega = \Phi/\Phi_0$ —a direct consequence of  $F = \xi\Phi^2$ —closes the tree-level  $\chi\psi\bar{\psi}$  and  $\chi \rightarrow h\bar{h}$  vertices exactly, forcing reheating through the conformal-anomaly channel.
- The discrete  $\mathbb{Z}_2^\psi$  symmetry (under an assumed anomaly-free UV gauge completion), together with the vanishing  $\chi\psi\bar{\psi}$  vertex, guarantees  $\psi$  stability; the condensate decays completely, leaving  $V_c$  as a frozen cosmological constant with  $w = -1$  (up to residual corrections  $\leq 10^{-10}$  bounded in App. D5).
- RG analysis shows  $\beta_{\lambda_0} \propto \lambda_0^2 > 0$  and  $\Delta\xi \leq 10^{-6}$ , confirming vacuum stability and the absence of fine-tuned running.

### *Quantitative estimates*

(parametric derivations with controlled uncertainties):

- $N$  is defined as  $N = \ln(a_{\text{end}}/a_*)$  with explicit matching assumptions (14). Planck  $2\sigma$  selects  $N \approx 45$ –56; anomaly-dominated  $T_{\text{reh}} \sim 10^9$  GeV maps to  $N \simeq 51$  in this convention (self-consistent  $T_{\text{reh}}^*$  for  $N = 50$  is  $\sim 4 \times 10^7$  GeV).

- Throughout the Planck  $2\sigma$  band, exact slow roll gives  $r \leq 0.0052$  at  $\xi = 11.1$ ; LiteBIRD  $r > 0.01$  would exclude the class.
- The dark-matter mass is fixed by the abundance matching at  $m_\psi \simeq 4.6 \times 10^{-3} H_{\text{inf}} \simeq 7.5 \times 10^{10} \text{ GeV}$  ( $g \simeq 1.0 \times 10^{-7}$ ) and follows from the gravitational-production mechanism; the absolute normalization requires a dedicated Floquet/lattice simulation.

### *Open questions*

(deferred to future work or UV completion):

- Inflationary initial conditions ( $\chi \gg M_{\text{Pl}}$ ) and the measure of the slow-roll attractor.
- The  $10^{110}$  hierarchy between  $V_0$  and  $V_c$  (the cosmological constant problem, shared with  $\Lambda\text{CDM}$ ).
- Absolute normalization of the dark-matter relic abundance (the abundance matching itself is fixed on the light branch, Sec. V) and the preheating energy partition (both requiring a lattice simulation).
- UV completion: anomaly cancellation for  $\mathbb{Z}_2^\psi$ , EFT cutoff and unitarity, swampland constraints.
- Baryogenesis and the Hubble tension (inherited from  $\Lambda\text{CDM}$  without resolution).

The key observational test is  $r$  under the locked  $N$  definition: Planck  $2\sigma$  allows  $N \approx 45\text{--}56$  with  $r \leq 0.0052$  (Table 1). A LiteBIRD-class measurement  $r > 0.01$  at 95% CL would exclude the model class in this convention. The frozen  $w_0 = -1$  prediction is falsifiable by DESI/Euclid at  $> 3\sigma$ , but it is shared with  $\Lambda\text{CDM}$ , so it discriminates neither model from the other; the sharp handles are the inflationary observables  $n_s$ ,  $r$ ,  $n_t$ ,  $f_{\text{NL}}$ .

### *Calibration vs. prediction (summary).*

Rigid predictions of the action at chosen  $(\xi, N)$  are the inflationary functions  $n_s(N)$  and  $r(N)$  and the derived scales  $H_{\text{inf}}$ ,  $m_\chi$ ,  $U^{1/4} = (3M_{\text{Pl}}^2 H_{\text{inf}}^2)^{1/4}$ . Calibrated or a-posteriori inputs are:  $\Omega_\Lambda$  (hence  $V_c$ ),  $\lambda_0$  (from  $A_s$ ),  $\xi$  (within the  $r$  bound), and the dark-matter Yukawa  $g$  (from  $\Omega_{\text{DM}}$ ; matched at  $g \simeq 1.0 \times 10^{-7}$  on the light branch, with the normalization pending the lattice computation). Dark energy is structurally present in the action but observationally degenerate with  $\Lambda\text{CDM}$ ; the hard falsification handle lives in the inflationary sector.

## Appendix A Numerical verification

The full CMB normalization proceeds in four steps:

### *Step 1: Plateau height from the potential.*

The Einstein-frame potential for  $\chi \gg M_{\text{Pl}}$  ( $\varphi \gg 1$ ) is

$$V_E(\chi) \simeq V_0 \left(1 - 2e^{-\beta\chi/M_{\text{Pl}}}\right), \quad V_0 \equiv \frac{\lambda_0 M_{\text{Pl}}^4}{4\xi^2}. \quad (\text{A1})$$

**Step 2: Slow-roll parameters and  $e$ -folds.**

With the potential slow-roll expressions of Eq. (22), the  $e$ -fold number between CMB horizon exit ( $\chi_*$ ) and the end of inflation ( $\epsilon_V = 1$  at  $\chi_{\text{end}}$ ) is

$$N \equiv \int_{\chi_*}^{\chi_{\text{end}}} \frac{H}{d\chi/dt} d\chi \simeq \frac{1}{M_{\text{Pl}}^2} \int_{\chi_*}^{\chi_{\text{end}}} \frac{V_E}{V'_E} d\chi = \frac{e^{\beta\chi_*/M_{\text{Pl}}}}{2\beta^2} - \frac{1}{2\beta^2} - \frac{\sqrt{2}}{\beta}, \quad (\text{A2})$$

where the last two terms are subleading for  $N \geq 50$ .

**Step 3: Scalar power spectrum.**

The standard single-field slow-roll formula gives

$$A_s = \left. \frac{V_E}{24\pi^2 M_{\text{Pl}}^4 \epsilon_V} \right|_{\chi=\chi_*} \simeq \frac{V_0}{24\pi^2 M_{\text{Pl}}^4} \frac{1}{2\beta^2 e^{-2\beta\chi_*/M_{\text{Pl}}}} = \frac{\lambda_0}{192\pi^2 \xi^2 \beta^2} e^{2\beta\chi_*/M_{\text{Pl}}}. \quad (\text{A3})$$

Using  $e^{\beta\chi_*/M_{\text{Pl}}} \simeq 2\beta^2 N$  from Step 2,

$$A_s \simeq \frac{\lambda_0 \beta^2 N^2}{48\pi^2 \xi^2} = \frac{\lambda_0 N^2}{12\pi^2 \xi^2 (6 + 1/\xi)}, \quad (\text{A4})$$

where we used  $\beta^2 = 4/(6 + 1/\xi)$  so that  $\beta^2/48 = 1/[12(6 + 1/\xi)]$ . Equation (A4) is algebraically identical to the result obtained directly from  $A_s = V_0/(24\pi^2 M_{\text{Pl}}^4 \epsilon_*)$  with  $\epsilon_* = \beta_o^2/(8N^2)$ ,  $\beta_o^2 = 6 + 1/\xi$ .

**Step 4: Solving for  $\lambda_0$ .**

Setting  $A_s = 2.1 \times 10^{-9}$  (Planck 2018 [16]) and  $N = 50$  gives the *large-field analytic* estimate

$$\lambda_0 = \frac{12\pi^2 \xi^2 (6 + 1/\xi) A_s}{N^2} \simeq 7.46 \times 10^{-8} \quad (\xi = 11.1, N = 50) \text{ (analytic; not the locked-} N \text{ table value)}. \quad (\text{A5})$$

*Note.* Earlier drafts printed the intermediate formula  $\lambda_0 \simeq 48\pi^2 \xi^2 A_s/N^2$  ( $4.90 \times 10^{-8}$  at  $\xi = 11.1$ ,  $N = 50$ ), which omits the factor  $(6 + 1/\xi)$  and is inconsistent with Eq. (A4). The large-field analytic estimate (A5) gives  $\lambda_0 \simeq 7.46 \times 10^{-8}$ ; the **locked- $N$  exact slow-roll inversion** used throughout the main text (Eqs. (15)–(16)) gives  $\lambda_0 \simeq 6.70 \times 10^{-8}$  at  $N = 50$  and  $\simeq 6.45 \times 10^{-8}$  at  $N = 51$ . Table 1 and all fiducial scales quote the locked- $N$  exact values.

The computational core is a one-dimensional root-find: at fixed  $N$  and  $\xi$ ,  $\lambda_0$  is obtained by requiring that the exact potential slow-roll amplitude  $A_s$  (Eqs. (15)–(16)) equals the observed  $2.100 \times 10^{-9}$ , with  $x_*$  determined by  $N = \ln(a_{\text{end}}/a_*)$  and  $\epsilon_V(x_{\text{end}}) = 1$ . The closed form quoted below is the *analytic large-field estimate* (A5) and is *not* the locked- $N$  value; the two differ because the analytic form neglects the  $\mathcal{O}(1)$  correction from evaluating  $\epsilon_V, \eta_V$  at the exact  $x_*(N)$ . The complete, executable implementation is `lock_n_convention.py` (together with

background\_and\_reheating.py); a ten-line listing cannot faithfully reproduce the locked value. For orientation:

```
import numpy as np
from scipy.optimize import brentq
xi = 11.1; M_P = 2.435e18; As_target = 2.1e-9
beta = 2.0/np.sqrt(6+1/xi) # paper convention
beta_o2 = 6+1/xi # our convention
N_fid = 50
# Analytic large-field estimate ONLY (Eq. A5): gives lam0 = 7.465e-08, NOT the locked value
lam0 = 12*np.pi**2*xi**2*beta_o2*As_target/N_fid**2
# Locked-N value requires inverting As at fixed N (root-find); results:
# N=50 -> lam0=6.70e-08, ns=0.9616, r=0.00425
# N=51 -> lam0=6.45e-08, ns=0.9623, r=0.00409; T_reh*(N=51)=1.4e9 GeV (Table I)
# H = sqrt(lam0)*M_P/(2*sqrt(3)*xi)
```

**Key results (locked  $N$ ,  $\xi = 11.1$ ).**

At  $N = 50$ :  $\lambda_0 = 6.70 \times 10^{-8}$ ,  $n_s = 0.9616$ ,  $r = 0.00425$ ,  $V_0 \simeq 4.78 \times 10^{63} \text{ GeV}^4$ ,  $H_{\text{inf}} \simeq 1.64 \times 10^{13} \text{ GeV}$ ,  $m_\chi \simeq 3.25 \times 10^{13} \text{ GeV}$ ,  $U^{1/4} \simeq 8.3 \times 10^{15} \text{ GeV}$ . At  $N = 51$  (anomaly  $T_{\text{reh}} \sim 10^9 \text{ GeV}$ ):  $\lambda_0 = 6.45 \times 10^{-8}$ ,  $n_s = 0.9623$ ,  $r = 0.00409$ ,  $H_{\text{inf}} \simeq 1.61 \times 10^{13} \text{ GeV}$ . In all cases  $U^{1/4} = (3M_{\text{Pl}}^2 H_{\text{inf}}^2)^{1/4}$ .

## Appendix B Conformal decoupling: screening and reheating

The structural reason the scalar decouples from the dark-matter fermion  $\psi$  and the Yukawa reheating channel closes is the exact proportionality  $\Omega = \Phi/\Phi_0$ , which follows from  $F(\Phi) = \xi\Phi^2$  and is specific to the induced-gravity coupling. (Solar-system fifth-force bounds on Standard Model matter are instead met by the large scalar mass  $m_\chi$ ; see Sec. II E.)

**Matter coupling ( $\psi$  sector).**

Under  $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$ ,  $\sqrt{-g} = \Omega^{-4} \sqrt{-\tilde{g}}$  and  $\psi = \Omega^{3/2} \tilde{\psi}$ . The Yukawa term transforms as

$$-g\Phi\bar{\psi}\psi\sqrt{-g} = -g\Phi_0\Omega \cdot \Omega^3 \cdot \Omega^{-4}\tilde{\bar{\psi}}\tilde{\psi}\sqrt{-\tilde{g}} = -g\Phi_0\tilde{\bar{\psi}}\tilde{\psi}\sqrt{-\tilde{g}}, \quad (\text{B6})$$

giving an exact,  $\chi$ -independent Einstein-frame mass  $m_E = g\Phi_0$  and a vanishing tree-level  $\chi\tilde{\bar{\psi}}\tilde{\psi}$  vertex. Static  $\psi$  matter in this sector does not source  $\chi$ , so  $\gamma = 1$  exactly within the  $\psi$  sector (observationally inert for solar-system tests, which probe SM matter; decisive for reheating and dark matter).

**Spin-connection cancellation.**

Under conformal transformation, the vierbein and Dirac matrices transform as  $\tilde{e}_\mu^a = \Omega e_\mu^a$ ,  $\tilde{\gamma}^\mu = \tilde{e}_a^\mu \gamma^a = \Omega^{-1} \gamma^\mu$ , and the spinor rescales as  $\psi = \Omega^{3/2} \tilde{\psi}$ . The massless Dirac kinetic term becomes

$$\begin{aligned} S_D &= \int d^4x \sqrt{-g} \bar{\psi} i \gamma^\mu (\partial_\mu + \frac{1}{4} \omega_\mu^{ab} [e] \gamma_{ab}) \psi \\ &= \int d^4x \sqrt{-\tilde{g}} \tilde{\bar{\psi}} i \tilde{\gamma}^\mu \left[ \partial_\mu \tilde{\psi} + \frac{3}{2} (\partial_\mu \ln \Omega) \tilde{\psi} + \frac{1}{4} \omega_\mu^{ab} [e] \gamma_{ab} \tilde{\psi} \right]. \end{aligned} \quad (\text{B7})$$

Using  $\omega_\mu^{ab}[e] = \omega_\mu^{ab}[\tilde{e}] - (e_\mu^a e_\nu^b - e_\mu^b e_\nu^a) \partial_\nu (\ln \Omega)$  and the identity  $\gamma^a \gamma_{ab} = 3\gamma_b$  in  $d = 4$ , the spin-connection contribution becomes

$$\frac{1}{4} \tilde{\gamma}^\mu \omega_\mu^{ab}[e] \gamma_{ab} = \frac{1}{4} \tilde{\gamma}^\mu \omega_\mu^{ab}[\tilde{e}] \gamma_{ab} - \frac{3}{2} \tilde{\gamma}^\nu \partial_\nu (\ln \Omega), \quad (\text{B8})$$

which exactly cancels the  $+\frac{3}{2}(\partial_\mu \ln \Omega)$  rescaling term, leaving  $S_D = \int d^4x \sqrt{-\tilde{g}} \tilde{\psi} i \tilde{\gamma}^\mu \tilde{\nabla}_\mu \tilde{\psi}$  with no derivative coupling to  $\chi$ .

### ***Reheating.***

The vanishing vertex means no perturbative  $\chi \rightarrow \tilde{\psi} \tilde{\psi}$  decay, for any  $g$ . For  $m_\psi \gtrsim m_\chi/2$  the decay is additionally kinematically forbidden; at the abundance-matched value  $g \simeq 1.0 \times 10^{-7}$  of Sec. V the kinematic closure no longer holds but the vertex closure still does. The time-varying Jordan-frame mass  $m_\psi = g|\Phi|$  is adiabatic for  $g \gtrsim 10^{-4}$  and non-adiabatic at the dark-matter value  $g \simeq 1.0 \times 10^{-7}$ ; this is the Jordan-frame image of the Einstein-frame gravitational production that sources the dark matter of Sec. V (a single physical process, not an additive channel), and it does not use the perturbative vertex. Reheating proceeds through conformal-anomaly production together with condensate decay (the gravitational pulse provides only a seed bath and cannot reheat on its own); under the locked  $N$  definition this selects  $N \simeq 45\text{--}56$  by Planck  $2\sigma$ , with anomaly  $T_{\text{reh}} \sim 10^9 \text{ GeV}$  at  $N \simeq 51$  (Sec. III, Table 1).

## **Appendix C Key derivations**

### ***1. Variational derivation of Eq. (3).***

$\delta S / \delta g_{\mu\nu} = 0$  with  $\delta(\sqrt{-g} \frac{1}{2} F R) = \frac{1}{2} F \sqrt{-g} G_{\mu\nu} \delta g^{\mu\nu} - \frac{1}{2} \sqrt{-g} (\nabla_\mu \nabla_\nu F - g_{\mu\nu} \square F) \delta g^{\mu\nu}$ . Adding kinetic and potential variations yields Eq. (3).

### ***2. Conformal transformation.***

$\Omega^2 = \xi \Phi^2 / M_{\text{Pl}}^2$ ,  $\tilde{R} = \Omega^{-2} (R - 6\Omega^{-1} \square \Omega)$ , producing  $d\chi/d\varphi = M_{\text{Pl}} \sqrt{6+1/\xi}$  and  $V_E = V_J / \Omega^4$  (Eq. (8)).

### ***3. Linearized field equation.***

Linearizing  $\square \Phi - V'(\Phi) + \xi \Phi R = 0$  with  $\Phi = \Phi_0 + \delta\Phi$  and  $V''(\Phi_0) = 2\lambda_0 \Phi_0^2$  yields Eq. (36); the quasi-static limit ( $\delta\ddot{\Phi} \ll H\delta\dot{\Phi} \ll m_{\text{eff}}^2 \delta\Phi$ ) gives Eq. (37). For  $m_\chi \gg H$  at all epochs after reheating (the ratio  $m_\chi/H$  never drops below  $10^{50}$ ), the quasi-static condition is satisfied to extreme precision, justifying the neglect of time-derivative terms.

### ***4. Junction-condition integration.***

Integrating Eq. (3) across the shell gives the scalar-tensor Lanczos condition (47). At  $\Phi_s = 0$ ,  $F(0) = F'(0) = 0$  and the condition degenerates.

### 5. Divergent $\sigma_E$ limit.

$\sigma_E = \int \sqrt{2V_E} (d\chi/d\varphi) d\varphi \sim \int e^{-2\varphi} d\varphi$  diverges at  $\varphi \rightarrow -\infty$  ( $\Phi = 0$ ), a conformal-mapping artifact.

## Appendix D Post-inflationary field dynamics and the dark-energy equation of state

This appendix supplies the rigorous derivation of the dark-energy results quoted in Sec. VI: the harmonic-minimum approximation, the coherent-oscillation and condensate-decay regimes, the overclosure bound, and the complete residual-motion budget (three sources, of which the residual-condensate tail is not manifest in the main text).

### 0. Numerical validation of the end-of-inflation state.

The quantities used in Secs. III–V are obtained by integrating the exact Klein–Gordon equation in the Einstein frame,  $\ddot{\chi} + 3H\dot{\chi} + V'_E(\chi) = 0$  with  $3M_{\text{Pl}}^2 H^2 = \frac{1}{2}\dot{\chi}^2 + V_E$ , rather than by evaluating slow-roll closed forms. Two conventions matter and are stated explicitly here. (i) In  $e$ -fold variables the Friedmann constraint reads  $H^2 = V_E/(3 - p^2/2)$  with  $p \equiv d\chi/dN$ , which degenerates as  $V_E \rightarrow 0$ ; we therefore integrate the  $e$ -fold system only over the slow-roll regime and switch to cosmic time for the oscillatory phase. (ii) The end of inflation is defined by  $\epsilon_V = 1$ , where slow roll has already broken down; the slow-roll velocity  $\dot{\chi} = -V'_E/(3H)$  must *not* be imposed there, since it would force  $K_{\text{end}} = V_{\text{end}}/3$  and  $\epsilon_H = 3/4$  by construction. Integrating instead from  $\epsilon_V = 0.05$  up to  $\epsilon_V = 1$  gives  $x_{\text{end}} = 0.76367$ ,  $\epsilon_H(x_{\text{end}}) = 0.4988$ ,  $V_{\text{end}}/V_0 = 0.28520$ ,  $K_{\text{end}}/V_{\text{end}} = 0.1994$ , and  $\rho_{\text{end}}/V_{\text{end}} = 1.1994$ . These reproduce the locked- $N$  fiducial values quoted in Sec. III and give the barrier ratio  $K_{\text{end}}/\Delta V_J \simeq 0.057$  of Sec. III. The same integration supplies the reheating ceiling used in Sec. III:  $T_{\text{reh}}^{\text{inst}}(N) = (30\rho_{\text{end}}(N)/\pi^2 g_*)^{1/4}$  decreases only as  $N^{-1/2}$ , whereas the tabulated  $T_{\text{reh}}^*(N)$  required by cosmological matching grows as  $e^{3.05N}$ ; the two intersect at  $N_{\text{max}} = 55.9$  (or 56.1 if the naive ceiling  $V_{\text{end}}^{1/4}$  is used instead). This is the reason the  $n_s$ -allowed band is quoted as  $N \approx 45$ –56:  $N = 57$  and  $N = 58$  would require a reheating temperature above instantaneous reheating. The numerical results quoted above and in Secs. III–VI were obtained with the analysis scripts accompanying this work: `background_and_reheating.py` (exact Klein–Gordon integration, reheating channels and the  $N$  window), `psi_production_bogoliubov.py` (de Sitter Bogoliubov index and the abundance matching), `psi_abundance_oscillating.py` (mode-equation abundance across the end-of-inflation transition), `dm_gap_closure_test.py` (abundance matching on the light branch with the exact mode-equation spectrum across the end-of-inflation transition, and the free-streaming check), and `residual_quintessence.py` (two-fluid Boltzmann integration and the  $\Delta w$  budget). These are available from the author on reasonable request.

### 1. Equation of motion near the minimum.

After conformal transformation to the Einstein frame, the canonical field  $\chi$  obeys

$$\ddot{\chi} + 3H\dot{\chi} + V_E'(\chi) = 0, \quad (\text{D9})$$

with

$$V_E(\chi) = V_c + \frac{1}{2}m_\chi^2\chi^2 + \mathcal{O}\left(\frac{m_\chi^2\chi^3}{M_{\text{Pl}}}\right), \quad (\text{D10})$$

where  $m_\chi^2 \equiv 2\lambda_0 M_{\text{Pl}}^2/[\xi^2(6+1/\xi)] \approx (3.25 \times 10^{13} \text{ GeV})^2$  at locked  $N = 50$  (Eq. (12)). The cubic term is *not* negligible at the largest post-inflationary amplitudes. Writing  $V_E = V_0(1 - e^{-x})^2$  with  $x \equiv \beta\chi/M_{\text{Pl}}$  and expanding about  $x = 0$  gives  $V_E = V_0[x^2 - x^3 + \mathcal{O}(x^4)]$ , so the cubic-to-quadratic ratio is  $\beta\chi/M_{\text{Pl}}$ —equal to  $\beta \simeq 0.81$  at  $\chi = M_{\text{Pl}}$ , and therefore  $\mathcal{O}(1)$  rather than  $\mathcal{O}(\lambda_0^{1/2})$ . The cubic coefficient is

$$c_3 = -\frac{V_0\beta^3}{M_{\text{Pl}}^3} = -\frac{\beta m_\chi^2}{2M_{\text{Pl}}} \simeq -1.8 \times 10^8 \text{ GeV}, \quad (\text{D11})$$

in the convention  $V \supset c_3\chi^3$ . The potential is thus effectively harmonic only for  $\chi \ll M_{\text{Pl}}/\beta \simeq 1.2 M_{\text{Pl}}$ ; at the end of inflation,  $\chi_{\text{end}} \simeq 0.94 M_{\text{Pl}}$ , the anharmonicity is  $\mathcal{O}(1)$ . This is why the post-inflationary equation of state is not exactly  $w = 0$  during the first oscillations—the exact integration of Sec. D0 gives  $\langle w \rangle \simeq -0.02$  over the first  $e$ -folds, approaching zero thereafter—and it is the origin of the  $\mathcal{O}(1)$  uncertainty in the mapping between  $N$  and  $T_{\text{reh}}$  noted in Sec. III. All numerical results in this work use the exact potential  $V_E$ , not the harmonic approximation. The ratio  $m_\chi/H_0 \approx 2.3 \times 10^{55}$  implies that any displacement from  $\chi = 0$  oscillates with frequency  $\gg H$  and is damped to zero within  $\sim m_\chi^{-1}$ .

### 2. Coherent oscillation and the dust redshift law.

Immediately after the end of slow-roll ( $\chi \sim M_{\text{Pl}}$ ,  $\epsilon \sim 1$ ), the field begins coherent oscillations around  $\chi = 0$ . For  $H \ll m_\chi$ , Eq. (D9) reduces to a damped harmonic oscillator with WKB solution

$$\chi(t) = \chi_0(t) \cos\left(\int^t m_\chi dt' + \theta_0\right), \quad \chi_0(a) = \chi_{\text{end}} \left(\frac{a}{a_{\text{end}}}\right)^{-3/2}, \quad (\text{D12})$$

which follows from the conserved adiabatic invariant  $I \equiv \oint p dq = \pi m_\chi \chi_0^2 a^3$ . The energy density of the oscillating condensate therefore redshifts as pressureless dust,

$$\rho_{\text{cond}}(a) = \frac{1}{2}m_\chi^2\chi_0^2(a) \propto a^{-3}, \quad (\text{D13})$$

which is the rigorous origin of the  $\rho \propto a^{-3}$  behavior quoted in Sec. IV.

### 3. Condensate decay: the overclosure bound.

The condensate must transfer its energy to SM radiation with high efficiency to avoid overclosing the universe. The observational constraint is

$$\rho_{\text{cond}}(a_{\text{reh}}) < \rho_{\text{DM}}^{(0)} \left( \frac{a_0}{a_{\text{reh}}} \right)^{-3} = \rho_{\text{DM}}^{(0)} \left( \frac{T_{\text{reh}}}{T_0} \right)^3, \quad (\text{D14})$$

where  $\rho_{\text{DM}}^{(0)} = \Omega_{\text{DM}} \rho_{\text{crit}}^{(0)} \approx 9.85 \times 10^{-48} \text{ GeV}^4$  and  $T_0 \approx 2.36 \times 10^{-13} \text{ GeV}$ . For  $T_{\text{reh}} \in [10^{-2}, 10^{15}] \text{ GeV}$  this gives  $\rho_{\text{cond}}(a_{\text{reh}}) < (5.2 \times 10^{-16} - 5.2 \times 10^{35}) \text{ GeV}^4$ . Comparing with the initial condensate energy density  $\rho_{\text{cond}}(a_{\text{end}}) \sim \frac{1}{2} m_\chi^2 M_{\text{Pl}}^2 \simeq 3 \times 10^{63} \text{ GeV}^4$ , the required suppression factor is  $\sim 2 \times 10^{26} - 2 \times 10^{77}$  (the upper end corresponding to  $T_{\text{reh}}$  at the BBN floor). This suppression is supplied by the decay of the condensate (channel (iv) of Sec. IV): because  $\Gamma t_0 \gg 1$  for any  $T_{\text{reh}} \gtrsim 10^{-2} \text{ GeV}$ , the residual condensate at  $a_0$  is strictly zero (D5c), and the locked  $N$  is confined to the Planck-compatible band (anomaly  $T_{\text{reh}} \sim 10^9 \text{ GeV} \Rightarrow N \simeq 51$ ).

### 4. Freezing at the minimum.

After reheating, the field sits at  $\chi \approx 0$  with precision set by the residual amplitude. Since  $m_\chi \gg H$  at all later epochs (including today, where  $m_\chi/H_0 \sim 10^{55}$ ), any small residual oscillation is damped to  $\chi = 0$  within

$$\sim m_\chi^{-1} \sim 2 \times 10^{-14} \text{ GeV}^{-1} \sim 10^{-38} \text{ s}, \quad (\text{D15})$$

effectively instantaneous compared to the Hubble time. With  $\chi \equiv 0$  to extraordinary precision, the Einstein-frame scalar energy-momentum tensor reduces to  $T_{\mu\nu}^{(\chi)} \rightarrow -V_c g_{\mu\nu}$ , the energy-momentum tensor of a cosmological constant; the equation of state  $w_0 = -1$  for all  $z$  then follows directly (Eq. (34)).

### 5. Complete residual-motion budget.

Sec. VI quotes the residual-motion bound  $\Delta w \ll 10^{-3}$ . Here we bound all three independent sources of residual  $\chi$  motion that could perturb the frozen minimum.

*a. Ricci driving.* In the Jordan frame, the linearized equation (Eq. (36)) has a source term  $\xi \Phi_0 R$  that displaces  $\Phi$  from  $\Phi_0$ :

$$\delta \Phi_{\text{qs}} \approx \frac{\xi \Phi_0 R}{2 \lambda_0 \Phi_0^2}. \quad (\text{D16})$$

During matter domination  $R \approx -3H^2(1-3w) \approx +3H^2$  for  $w = 0$ , so

$$\frac{\delta \Phi_{\text{qs}}}{\Phi_0} \sim \frac{\xi H^2}{\lambda_0 M_{\text{Pl}}^2} \sim \left( \frac{H}{m_\chi} \right)^2. \quad (\text{D17})$$

At  $z = 0$  ( $H = H_0$ ):  $\delta\Phi/\Phi_0 \sim 10^{-111}$ . Varying  $\Phi$  produces kinetic energy  $K \sim \frac{1}{2}\dot{\delta\Phi}^2 \sim \frac{1}{2}(\delta\Phi \cdot H)^2$ , giving

$$\Delta w_{\text{Ricci}} \equiv \frac{K}{V_c} \sim \left(\frac{H_0}{m_\chi}\right)^2 \sim 2 \times 10^{-111}, \quad (\text{D18})$$

$10^{108}$  orders below the DESI  $3\sigma$  sensitivity [35].

*b. Quantum fluctuations.* For a massive scalar in de Sitter space with  $m \gg H$ ,  $\langle\chi^2\rangle \approx 3H^4/(8\pi^2 m_\chi^2)$ . The corresponding energy density is  $\rho_{\text{quant}} \sim 3H_0^4/(16\pi^2) \ll V_c$ , yielding

$$\Delta w_{\text{quant}} \leq H_0^4/V_c \sim 2 \times 10^{-121}. \quad (\text{D19})$$

*c. Residual condensate tail.* This third source—not manifest in the main-text summary of Sec. VI—is the residual condensate that survives reheating. The condensate is tracked by the coupled two-fluid system  $d(\rho_{\text{cond}}a^3)/dN = -(\Gamma/H)\rho_{\text{cond}}a^3$  and  $d(\rho_{\text{rad}}a^4)/dN = +(\Gamma/H)\rho_{\text{cond}}a^4$ , integrated from  $a_{\text{end}}$  to  $a_0$  with  $H^2 = (\rho_{\text{cond}} + \rho_{\text{rad}})/(3M_{\text{Pl}}^2)$  and an exponential integrator for the decay term (the ratio  $\Gamma/H$  reaches  $\sim 10^{30}$  at late times). The result is that for every  $\Gamma \neq 0$  the residual today underflows to zero: the decay-completeness condition  $\Gamma t_0 \gg 1$  holds by  $\geq 20$  orders of magnitude across  $T_{\text{reh}} \in [10^{-2}, 10^{15}]$  GeV. The overclosure bound is therefore not a constraint that happens to be met but a statement about what the residual *would* be for a stable condensate; that hypothetical value,

$$\rho_{\text{cond}}(a_0)|_{\Gamma=0} = \rho_{\text{end}}(a_{\text{end}}/a_0)^3 \simeq 1.65 \times 10^{-29} \text{ GeV}^4, \quad (\text{D20})$$

exceeds  $\rho_{\text{DM}}^{(0)}$  by  $\sim 10^{18}$  and would overclose the universe. With the decay included the upper bound

$$\rho_{\text{cond}}(a_0) < \rho_{\text{DM}}^{(0)} \approx 1.06 \times 10^{-47} \text{ GeV}^4, \quad (\text{D21})$$

giving the constraint  $\rho_{\text{cond}}(a_0)/V_c \leq \Omega_{\text{DM}}/\Omega_\Lambda \sim 0.39$ . This is a *requirement*, not a computed value: an absolutely stable condensate would give  $\rho_{\text{cond}}(a_0)/V_c \simeq 7 \times 10^{17}$  and would overclose the universe by  $\sim 10^{18}$ . The requirement is met because the condensate decays: the decay-completeness condition  $\Gamma t_0 \gg 1$  holds by  $\geq 20$  orders of magnitude for any  $T_{\text{reh}} \gtrsim 10^{-2}$  GeV, so the residual at  $a_0$  is strictly zero (see D5c). This bounds the residual condensate’s contribution to the dark-energy equation of state; combined with the rapid damping (D7) it confirms that the condensate is cosmologically irrelevant at late times, as asserted in Sec. VI.

## 6. Summary.

The complete dark-energy sector is captured by

$$V_E(0) = V_c, \quad m_\chi^2 = \frac{2\lambda_0 M_{\text{Pl}}^2}{\xi^2(6 + 1/\xi)}, \quad w_0 = -1, \quad (\text{D22})$$

with

$$\begin{aligned} |\Delta w| &< (H_0/m_\chi)^2 \sim 2 \times 10^{-111} \text{ (Ricci)} \\ &+ \leq 2 \times 10^{-121} \text{ (quantum)} \\ &+ 0 \text{ (condensate tail; } \Gamma t_0 \gg 1), \end{aligned} \quad (\text{D23})$$

the third line vanishes because  $\Gamma t_0 \gg 1$  for any  $T_{\text{reh}} \gtrsim 10^{-2}$  GeV (see D5c), so the condensate tail contributes to  $\Omega_m$  but not to  $\Delta w_{\text{DE}}$ ; and  $\Delta \mathcal{A}/\mathcal{A} = +\xi R/(\lambda_0 \Phi_0^2)$ .

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***Modifications relative to earlier drafts of this manuscript.***

(1) The  $e$ -fold number is locked to  $N = \ln(a_{\text{end}}/a_*)$  with explicit matching assumptions; all tabulated  $n_s$ ,  $r$ ,  $\lambda_0$  use exact potential slow-roll at that  $N$  (not mixed attractor closed forms). (2) At  $\xi = 11.1$ :  $N = 50 \Rightarrow \lambda_0 = 6.70 \times 10^{-8}$ ,  $r = 0.00425$ ,  $n_s = 0.9616$ ; anomaly  $T_{\text{reh}} \sim 10^9$  GeV  $\Rightarrow N \simeq 51$ . Planck  $2\sigma$  band  $N \approx 45$ – $56$  has  $r \leq 0.0052$ . (3) Appendix no longer promotes the incorrect  $\lambda_0 = 48\pi^2 \xi^2 A_s/N^2$  formula. (4)  $\mathbb{Z}_2^\psi$  stability is conditional on an anomaly-free UV discrete gauge completion. (5) Dark energy is a calibrated frozen  $V_c$ ; single-field residual quintessence is excluded by  $m_\chi/H_0 \sim 10^{55}$ . (6) A parallel topological-defect DM channel is listed. (7) Figures regenerated with the locked- $N$  convention (`scripts/lock_n_convention.py`). (8) The dark-matter production exponent is corrected from  $e^{-\pi m_\psi/H_{\text{inf}}}$  to the exact Bogoliubov result  $e^{-2\pi m_\psi/H_{\text{inf}}}$ ; the exponential-framework matching point is  $g \simeq 6.9 \times 10^{-5}$ . (9) The gravitational pulse of channel (i) does not reheat the universe (radiation fraction scales as  $a^{-1}$ ); reheating is attributed solely to the conformal-anomaly channel. (10)  $K_{\text{end}} = \epsilon_H V_{\text{end}}/(3 - \epsilon_H) \simeq 0.199 V_{\text{end}}$ ; the Planck band becomes  $N \approx 45$ – $56$  ( $N_{\text{max}} = 55.9$ ). (11)  $\Delta w_{\text{quant}} \lesssim 2 \times 10^{-121}$ ; the residual-condensate bound  $\leq 0.39$  is a constraint satisfied because  $\Gamma t_0 \gg 1$ . (12) The QCD  $b_3$  and  $\alpha_s(m_\chi)$  entering the anomaly are given explicitly. (13) Appendix D records the end-of-inflation numerical validation. (14) Abundance matching with the true mode-equation spectrum selects the light branch  $g \simeq 1.0 \times 10^{-7}$  ( $m_\psi \simeq 7.5 \times 10^{10}$  GeV); the heavy exponential-framework point overproduces at the model’s own  $T_{\text{reh}}$ ; absolute normalization awaits a lattice computation.

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- **Code availability:** Analysis scripts are available from the author on reasonable request.
- **Author contributions:** Yaming Hu performed all research, calculations, and writing for this work.

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